BookLeaf Author Support and Communication Portal

Software Design Document

Version	1.1: adds author self-registration, the fixed-admin model, the production seed guard and the synthetic-data cleanup tool (matches branch `integration/bookleaf-fullstack`)
Date	2026-10-16
Status	Describes the implemented application. Items not implemented or not verified are stated as such. The registration release is implemented and tested locally but **not yet deployed**; production still runs the previous release
Repository	private GitHub repository `singhdaksh7/book-leaf__assignment-1`
Staging	https://bookleaf.77.37.45.177.sslip.io (runs API `3eb1aff` and web `12edd72` with the real OpenRouter provider and self-registration; the assignment's 10 authors and 18 books are imported (358 field comparisons against the original JSON, 0 mismatches) alongside older synthetic test records, which are inventoried but not deleted; the free daily AI quota was exhausted on 2026-10-16, so live AI analyses end with a visible rate-limit notice, see §16 and §21)

Reading note. Every statement describes code that exists in the repository. Where a capability was tested only with mocks, or not tested, the text says so. Detailed per-component documents are in `docs/architecture/`; this document summarises them and links the diagrams.

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1. Project overview and objectives

BookLeaf Publishing supports authors whose books are in production or published. The portal lets an author raise a support ticket about a specific book and follow the conversation, and lets an administrator triage the queue, discuss and reply, with an AI assistant that proposes a category, a priority and a draft reply.

Objectives:

1. Give each author a private, authenticated view of their own books and tickets.
2. Let administrators work a queue with status, priority, assignment, replies and internal notes.
3. Use AI to speed up triage **without ever sending anything automatically**: a human reviews, edits and explicitly sends every reply.
4. Be secure by default, reproducible (Docker Compose), documented and tested.

2. Problem statement

Author questions (royalties, ISBNs, print quality, distribution, production status) arrive through informal channels, are hard to track, and answers depend on who reads them. Replies need consistency with company policy, and care that a response never promises an amount or date the company has not

confirmed. The portal centralises requests, keeps history, isolates authors from each other, and uses a policy-grounded AI draft to make administrators faster while keeping a human accountable.

3. Requirements

3.1 Functional requirements

ID	Requirement	Status
F1	Sign in and out with e-mail and password; roles AUTHOR and ADMIN	Implemented
F1a	Authors create their own account (public self-registration). Administrators are never self-registered: the single admin is created and managed by a server-side tool	Implemented
F2	Author lists and views only their own books	Implemented
F3	Author creates a ticket for one of their books (subject, description)	Implemented
F4	Author attaches files to a ticket and downloads them	Implemented
F5	Author and admin exchange replies on a ticket	Implemented
F6	Admin lists/filters/paginates the queue and opens a ticket	Implemented
F7	Admin changes status, category and priority; CLOSED is terminal	Implemented
F8	Admin writes internal notes invisible to the author	Implemented
F9	Admin assigns a ticket to themself	Implemented
F10	AI classifies each ticket (six categories, priority, confidence) and drafts a reply with cited policy	Implemented; real-model quality only partly verified (§17)
F11	Admin reviews, edits and explicitly sends a reply; AI never sends	Implemented
F12	Live updates (SSE) with polling fallback	Implemented
F13	Author dashboard with ticket counts	Implemented
—	Password reset, e-mail notifications, attachment deletion, server-side search	**Not implemented**

3.2 Non-functional requirements

Area	Requirement and how it is met
Security	Server-side sessions, CSRF, origin checks, Argon2id, RBAC, per-author scoping, input validation, rate limiting (§14, §20)
Privacy	AI receives minimised text: no e-mail, phone, name or financial figures (§16)
Reliability	AI failures never block ticket handling; bounded retries; single-flight attempts; health checks
Cost	AI restricted to zero-price models; explicit output caps and timeouts (§16)
Maintainability	Shared Zod contracts across API, AI and web; TypeScript throughout; layered tests (§23)

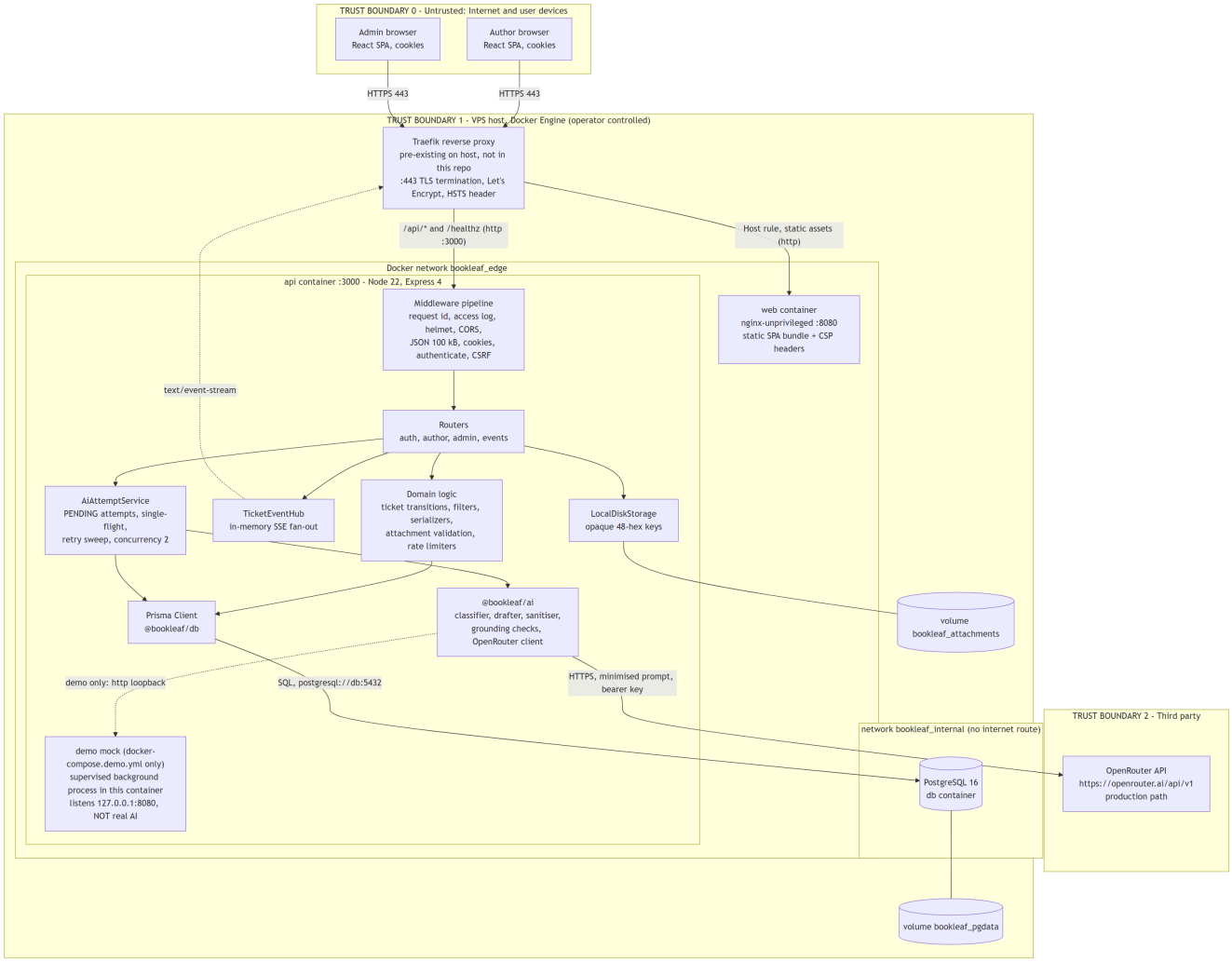
Area	Requirement and how it is met
Operability	Docker Compose, request ids, structured logs, backup scripts (§21, §22)
Performance	Not load-tested; no performance targets are claimed
Availability	Single host, single API instance; no redundancy

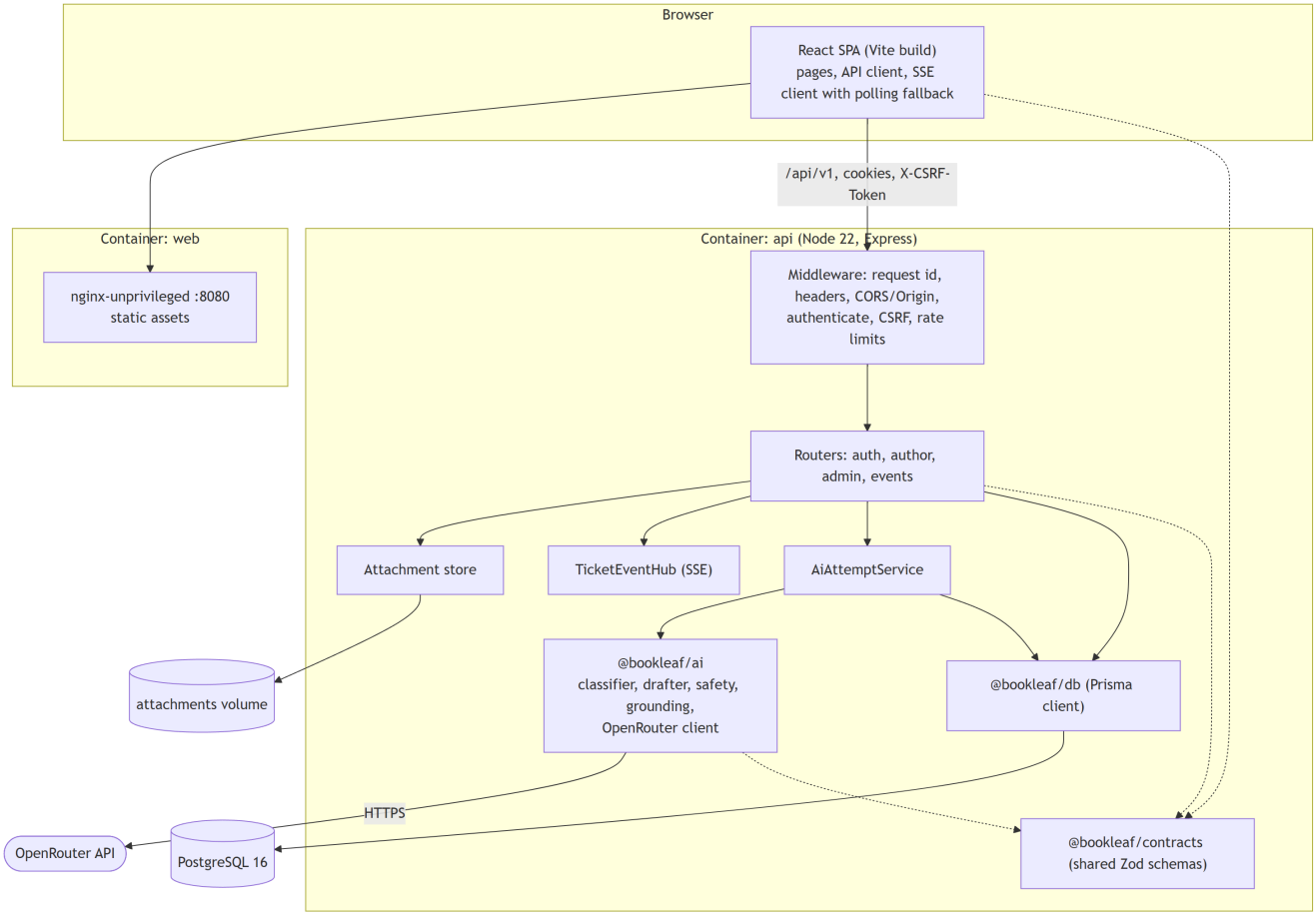
4. Technology stack

Layer	Technology
Frontend	React, Vite, TypeScript, React Router; served by nginx-unprivileged
Backend	Node.js 22, Express, TypeScript, Zod, Argon2id
Database	PostgreSQL 16, Prisma ORM, SQL migrations with triggers
AI	`@bookleaf/ai` package; OpenRouter chat-completions API (free models), built-in policy knowledge base
Shared	`@bookleaf/contracts` (Zod schemas and types)
Infrastructure	Docker Compose, shared Traefik (TLS via Let's Encrypt) on a VPS
Testing	Vitest, Supertest, Testing Library, Playwright, `node --test`
Documentation	Mermaid diagram sources, OpenAPI file

5. High-level system architecture

The system is three containers (web, api, db) behind a shared Traefik reverse proxy. The browser loads static assets from the web container and calls the API at `/api/v1` on the same origin. The API owns authentication, authorisation, business rules, the database, the attachment volume, the event stream and the AI integration. OpenRouter is the only external service the application calls.


TRUST BOUNDARY 0 - Untrusted: Internet and user devices
Admin browser
React SPA, cookies
Author browser
React SPA, cookies
HTTPS 443
HTTPS 443
TRUST BOUNDARY 1 - VPS host, Docker Engine (operator controlled)
Traefik reverse proxy
pre-existing on host, not in this repo
:443 TLS termination, Let's Encrypt, HSTS header
/api/* and /healthz (http :3000)
Host rule, static assets (http)
Docker network bookleaf_edge
api container :3000 - Node 22, Express 4
Middleware pipeline
request id, access log, helmet, CORS, JSON 100 kB, cookies, authenticate, CSRF
web container
nginx-unprivileged :8080
static SPA bundle + CSP headers
text/event-stream
Routers
auth, author, admin, events
AiAttemptService
PENDING attempts, single-flight, retry sweep, concurrency 2
TicketEventHub
In-memory SSE fan-out
Domain logic
ticket transitions, filters, serializers, attachment validation, rate limiters
LocalDiskStorage
opaque 48-hex keys
volume bookleaf_attachments
Prisma Client
@bookleaf/db
@bookleaf/ai
classifier, drafter, sanitiser, grounding checks, OpenRouter client
demo only: http loopback
SQL, postgresql://db:5432
HTTPS, minimised prompt, bearer key
demo mock (docker-compose.demo.yml only)
supervised background process in this container
listens 127.0.0.1:8080, NOT real AI
network bookleaf_internal (no internet route)
PostgreSQL 16
db container
TRUST BOUNDARY 2 - Third party
OpenRouter API
https://openrouter.ai/api/v1
production path
volume bookleaf_pgdata



Diagrams are embedded as high-resolution images (zoom in the PDF for detail); editable Mermaid sources and SVGs are in `docs/architecture/diagrams/`.

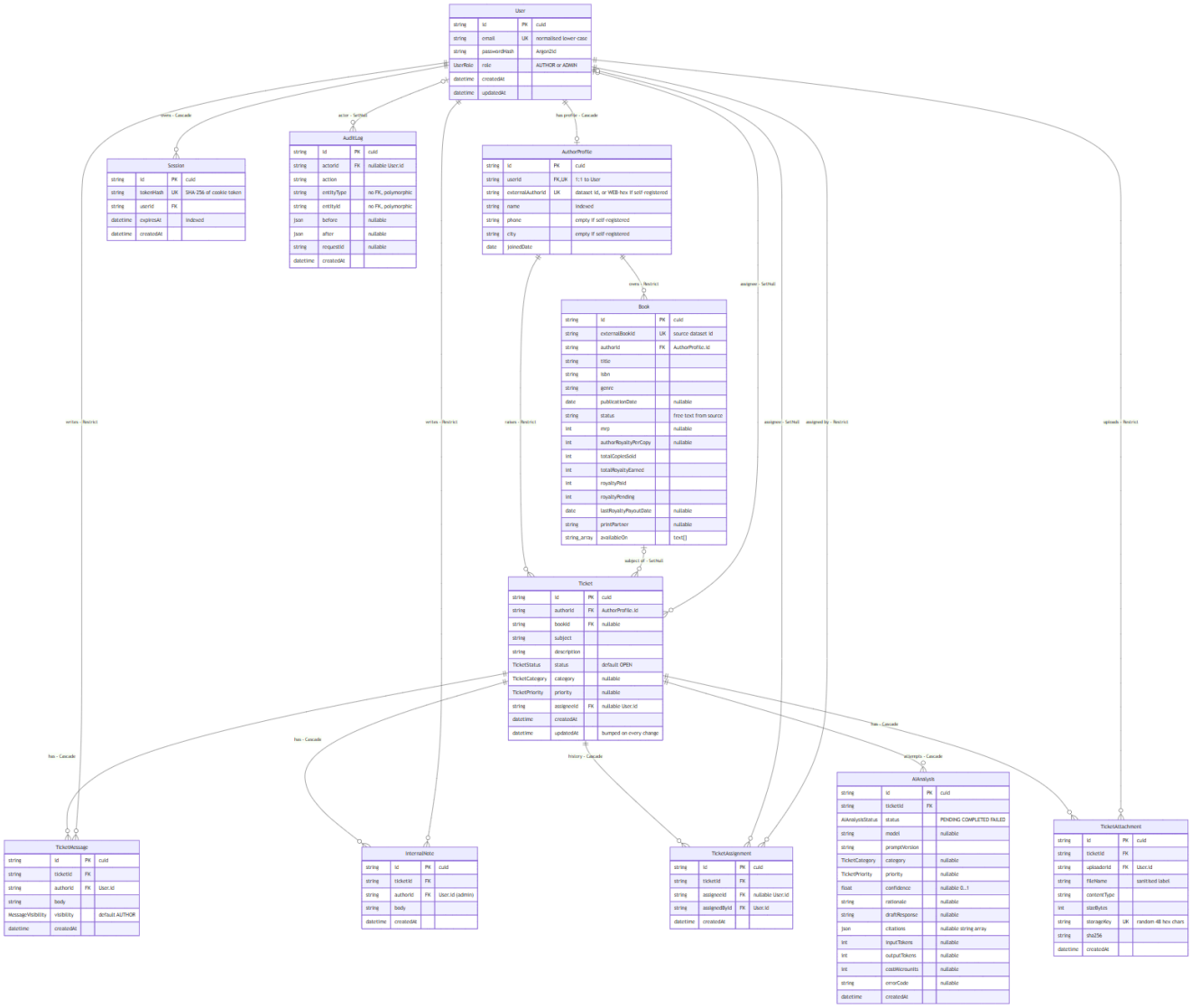
Monorepo layout: `apps/web`, `apps/api`, `packages/contracts`, `packages/db`, `packages/ai`, `infra`, `docs`, `tests`.

6. Database design

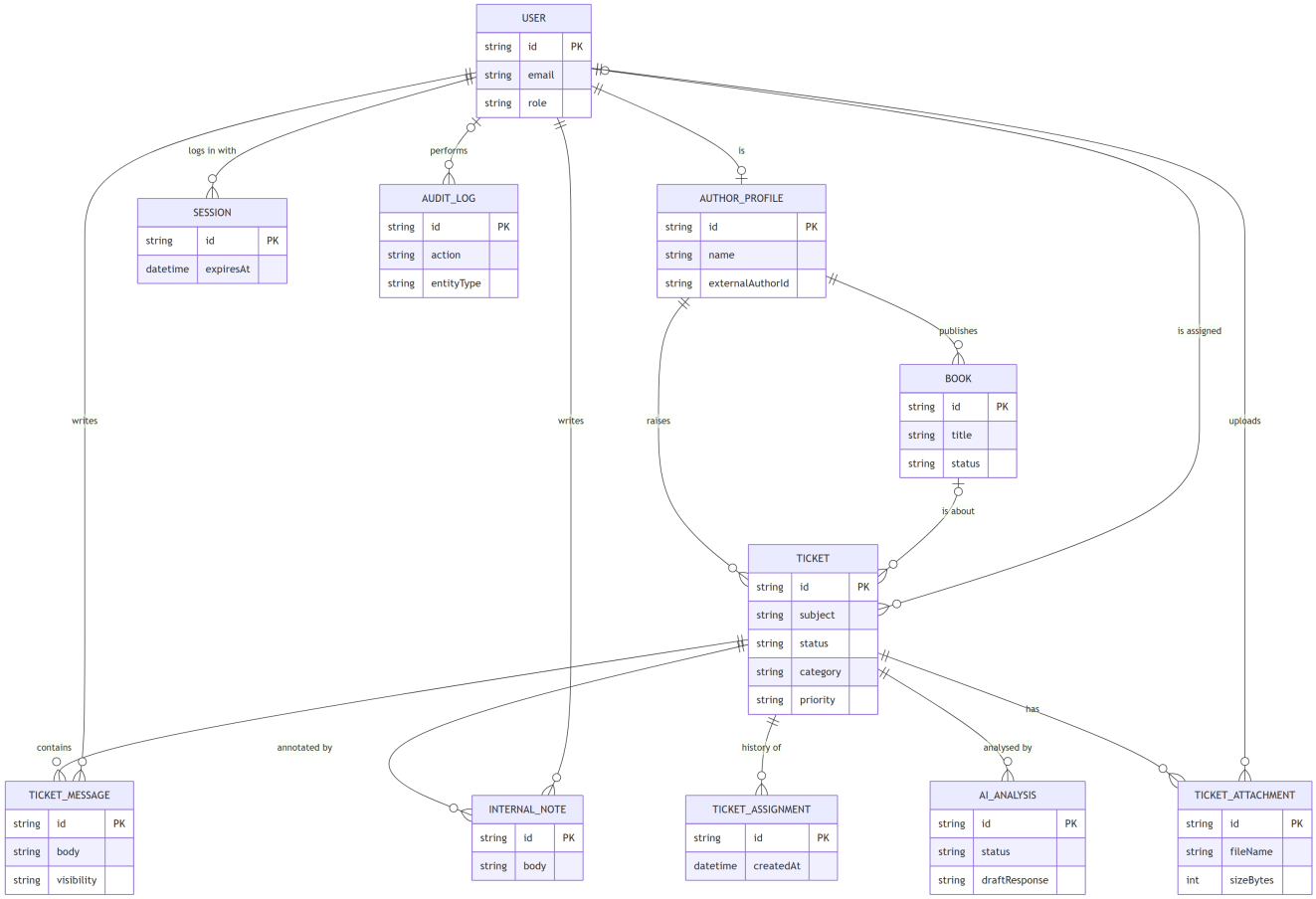
Eleven tables. `User` has an optional one-to-one `AuthorProfile`; an author owns `Book`s and opens `Ticket`s (optionally about a book). A ticket owns messages, internal notes, assignments, attachments, AI analysis attempts and audit entries. `Session` rows belong to users. Enumerations: `UserRole` (AUTHOR, ADMIN), `TicketStatus` (OPEN, IN_PROGRESS, RESOLVED, CLOSED), `TicketCategory` (ROYALTY_PAYMENTS, ISBN_METADATA, PRINTING_QUALITY, DISTRIBUTION_AVAILABILITY, BOOK_STATUS_PRODUCTION, GENERAL_INQUIRY), `TicketPriority` (CRITICAL, HIGH, MEDIUM, LOW), `MessageVisibility` (AUTHOR, ADMIN), `AiAnalysisStatus` (PENDING, COMPLETED, FAILED).

Database-level protections include a trigger requiring a ticket's book to belong to the ticket's author, a trigger requiring assignees to be ADMIN users, and a partial unique index allowing one PENDING AI attempt per ticket. Deleting internal notes or assignments is **not** trigger-protected (known limitation).

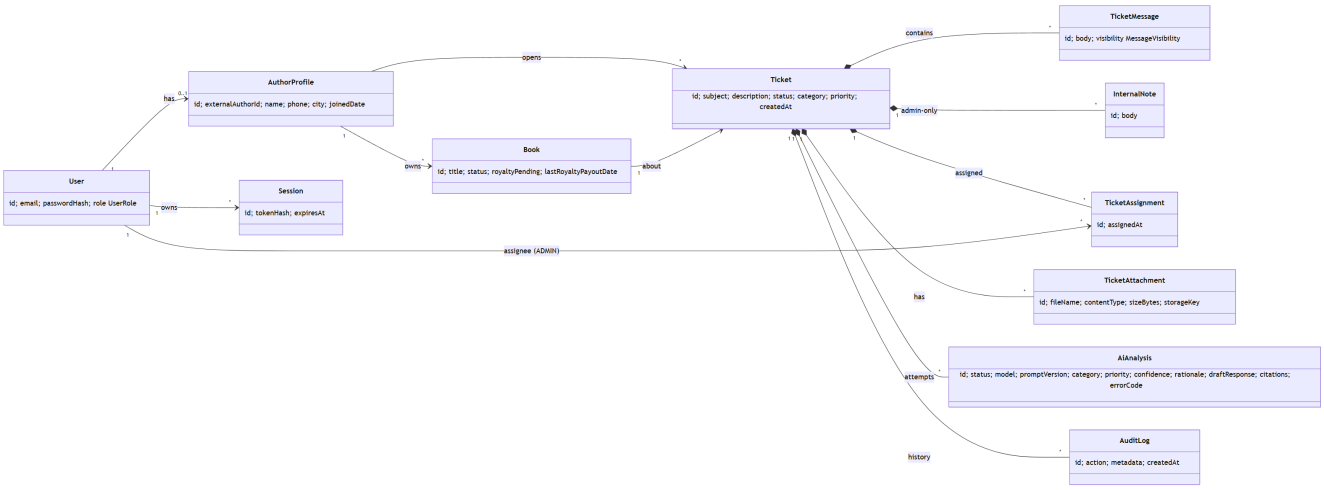
6.1 Full ERD


User
string id PK cuid
string email UK normalized lower-case
string passwordHash argon2id
UserRole role AUTHOR or ADMIN
datetime createdAt
datetime updatedAt
owns · Cascade
actor · SetNull
has profile · Cascade
Session
string id PK cuid
string tokenHash UK SHA-256 of cookie token
string userId FK
datetime expiresAt indexed
datetime createdAt
AuditLog
string id PK cuid
string actorId FK nullable User.id
string action
string entityType no FK, polymorphic
string entityId no FK, polymorphic
json before nullable
json after nullable
string requestId nullable
datetime createdAt
AuthorProfile
string id PK cuid
string userId FK,UK 1:1 to User
string externalAuthorId UK dataset id, or WEB-hex if self-registered
string name indexed
string phone empty if self-registered
string city empty if self-registered
date joinedDate
owns · Restrict
assignee · SetNull
Book
string id PK cuid
string externalBookId UK source dataset id
string authorId FK AuthorProfile.id
string title
string isbn
string genre
date publicationDate nullable
string status free text from source
int msrp nullable
int authorRoyaltyPerCopy nullable
int totalCopiesSold
int totalRoyaltyEarned
int royaltyPaid
int royaltyPending
date lastRoyaltyPayoutDate nullable
string printPartner nullable
string_array availableOn text[]
writes · Restrict
writes · Restrict
raises · Restrict
assignee · SetNull
assigned by · Restrict
uploads · Restrict
subject of · SetNull
Ticket
string id PK cuid
string authorId FK AuthorProfile.id
string bookId FK nullable
string subject
string description
TicketStatus status default OPEN
TicketCategory category nullable
TicketPriority priority nullable
string assigneeId FK nullable User.id
datetime createdAt
datetime updatedAt bumped on every change
has · Cascade
has · Cascade
history · Cascade
has · Cascade
attempts · Cascade
TicketMessage
string id PK cuid
string ticketId FK
string authorId FK User.id
string body
MessageVisibility visibility default AUTHOR
datetime createdAt
InternalNote
string id PK cuid
string ticketId FK
string authorId FK User.id (admin)
string body
datetime createdAt
TicketAssignment
string id PK cuid
string ticketId FK
string assigneeId FK nullable User.id
string assignedById FK User.id
datetime createdAt
AiAnalysis
string id PK cuid
string ticketId FK
AiAnalysisStatus status PENDING COMPLETED FAILED
string model nullable
string promptVersion
TicketCategory category nullable
TicketPriority priority nullable
float confidence nullable 0..1
string rationale nullable
string draftResponse nullable
json citations nullable string array
int inputTokens nullable
int outputTokens nullable
int costMicrounits nullable
string errorCode nullable
datetime createdAt
TicketAttachment
string id PK cuid
string ticketId FK
string uploaderId FK User.id
string fileName sanitised label
string contentType
int sizeBytes
string storageKey UK random 48 hex chars
string sha256
datetime createdAt

6.2 Simplified ERD



6.3 Domain class diagram



7. Data dictionary (summary)

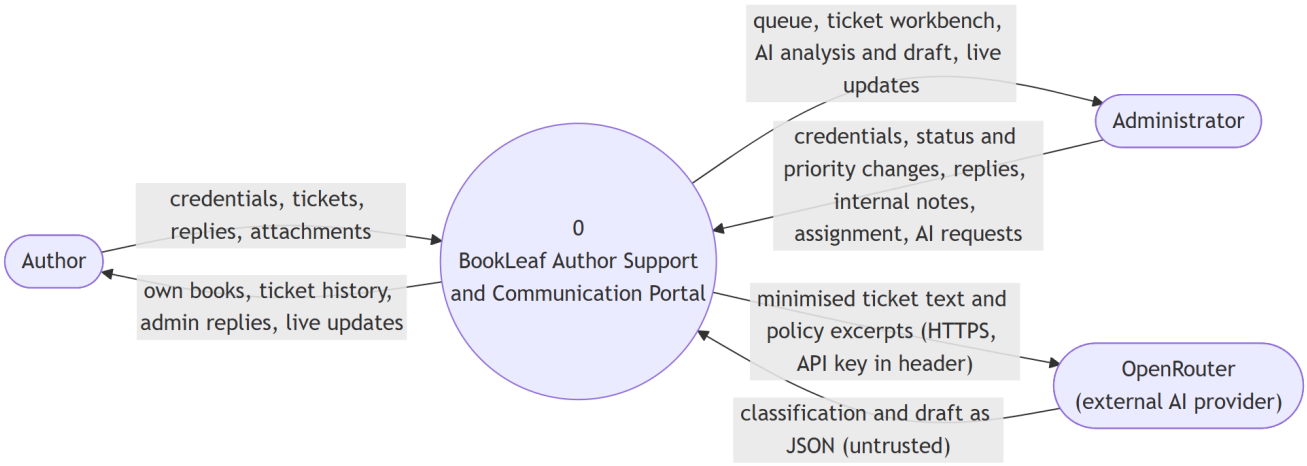
Full column-level dictionary with types, keys, defaults and delete behaviour: `docs/architecture/04-DATA-DICTIONARY.md`.

Table	Purpose	Key columns
User	Login identity	id, email (unique, lower-case), passwordHash (Argon2id), role

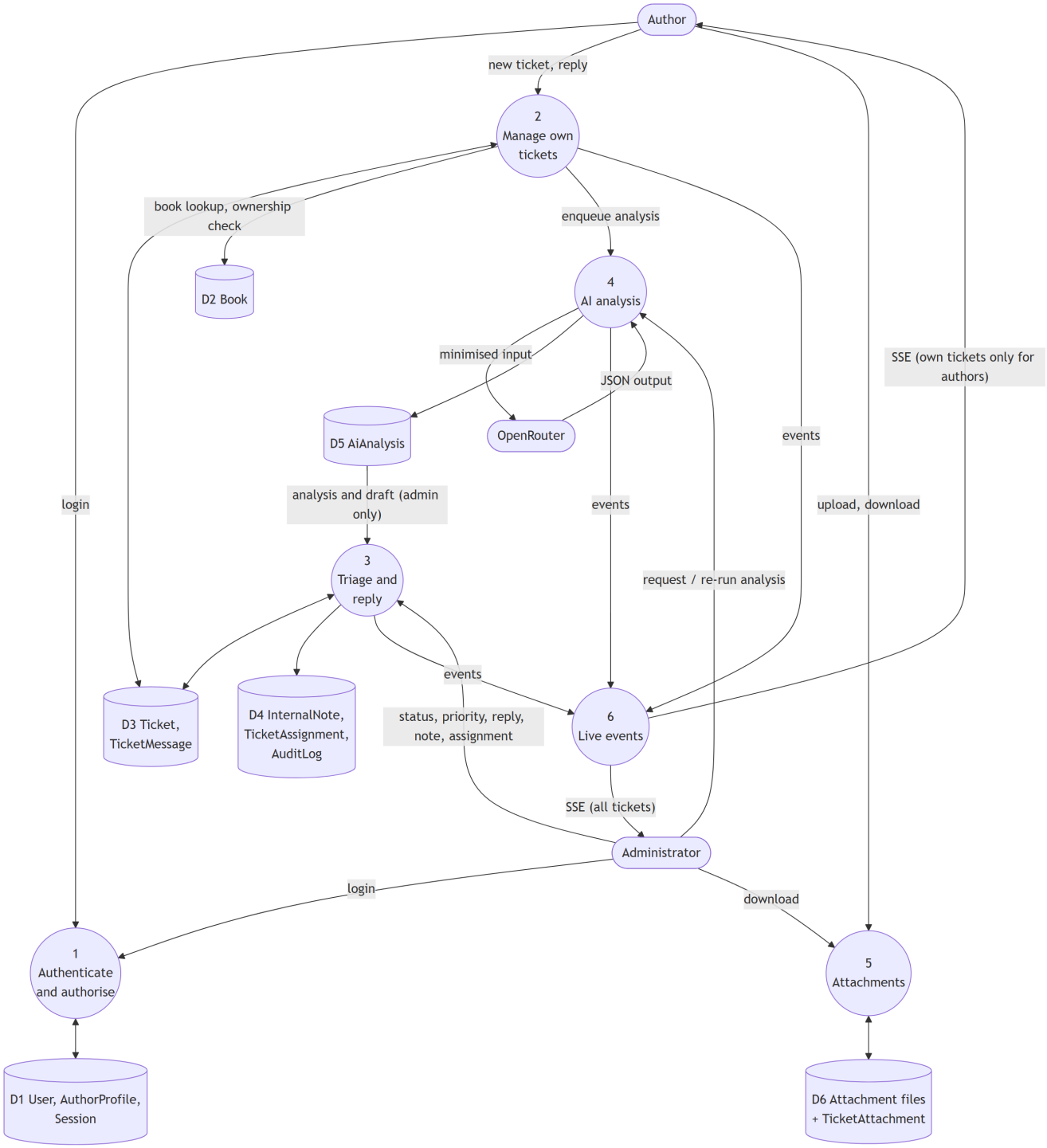
Table	Purpose	Key columns
AuthorProfile	Author details from the source dataset	id, userId (unique), externalAuthorId (unique), name, phone, city, joinedDate
Book	A book owned by an author	id, externalBookId, authorId, title, isbn, status, royalty figures, availableOn
Session	Server-side login session	id, tokenHash (SHA-256 of cookie token), userId, expiresAt
Ticket	A support request	id, authorId, bookId?, subject, description, status, category?, priority?, assigneeId?
TicketMessage	Conversation entry	ticketId, authorId, body, visibility
InternalNote	Admin-only note	ticketId, authorId, body
TicketAssignment	Assignment history	ticketId, assigneeId?, assignedById
TicketAttachment	Uploaded file metadata	ticketId, uploaderId, fileName, contentType, sizeBytes, storageKey (unique), sha256
AiAnalysis	One AI attempt	ticketId, status, model, promptVersion, category, priority, confidence, rationale, draftResponse, citations, token counts, costMicrounits, errorCode
AuditLog	Change history	actorId?, action, entityType, entityId, before, after, requestId

8. Data flow diagrams

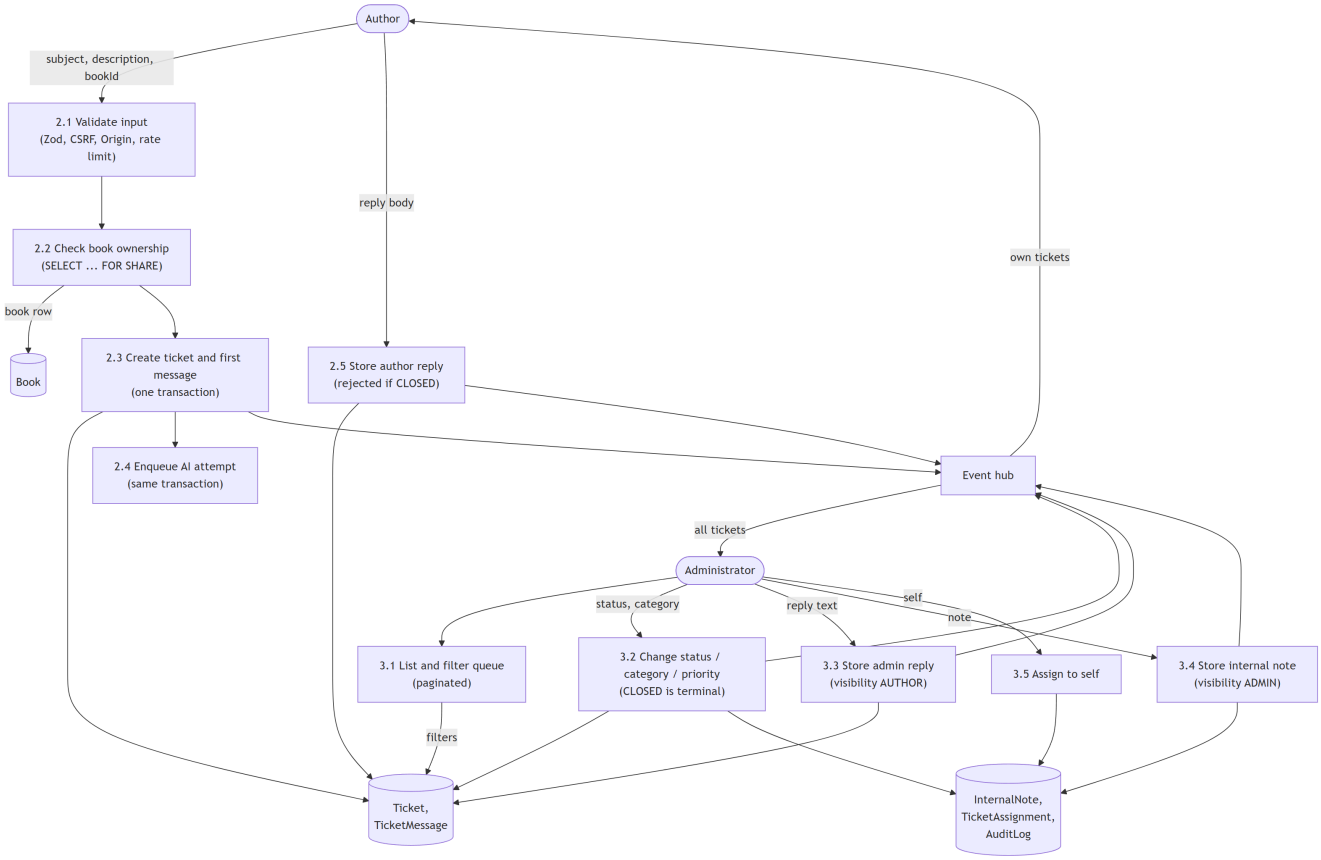
8.1 Level 0 (context)



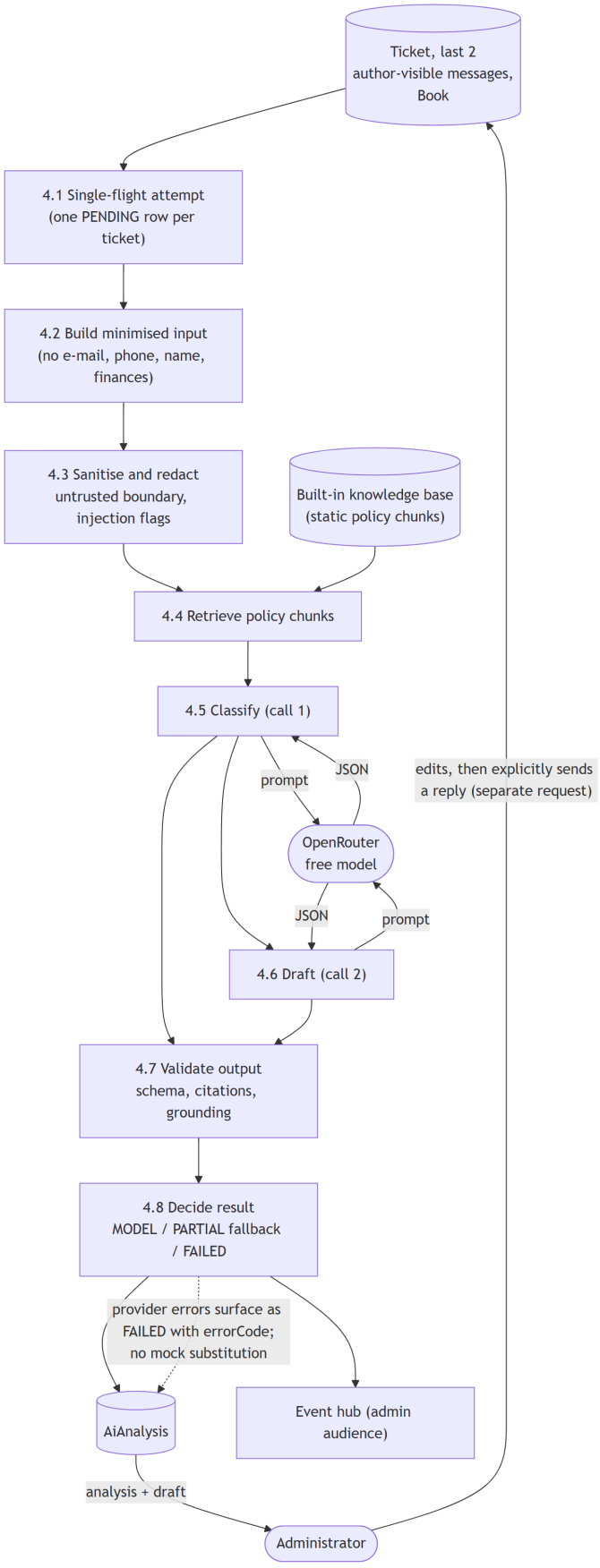
8.2 Level 1


Author
new ticket, reply
2
Manage own tickets
book lookup, ownership check
D2 Book
enqueue analysis
4
AI analysis
minimised input
JSON output
SSE (own tickets only for authors)
D5 AiAnalysis
OpenRouter
events
login
analysis and draft (admin only)
events
upload, download
3
Triage and reply
request / re-run analysis
events
D3 Ticket, TicketMessage
D4 InternalNote, TicketAssignment, AuditLog
status, priority, reply, note, assignment
6
Live events
SSE (all tickets)
Administrator
login
download
1
Authenticate and authorise
5
Attachments
D1 User, AuthorProfile, Session
D6 Attachment files + TicketAttachment

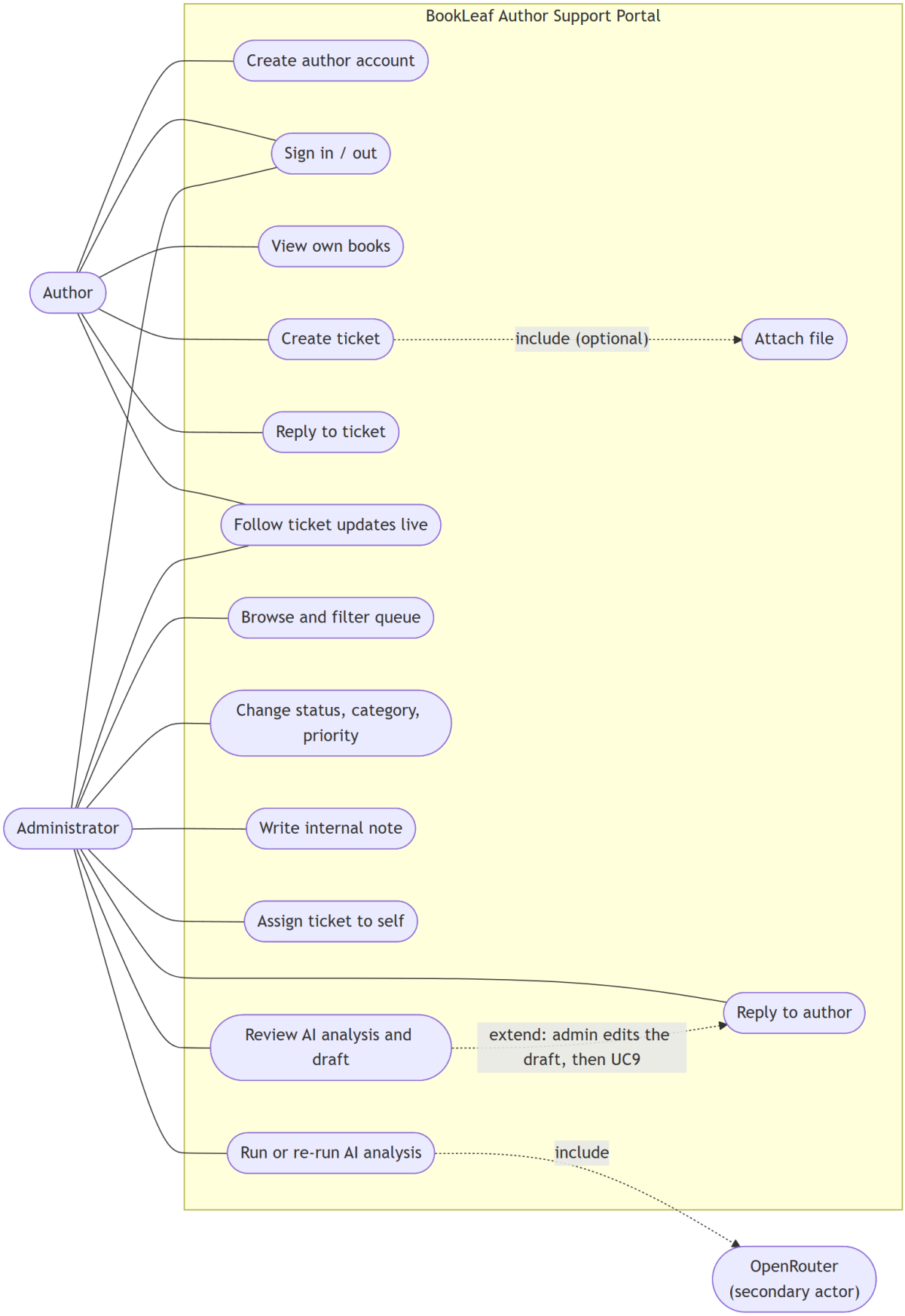
8.3 Level 2: ticket workflow



8.4 Level 2: AI workflow



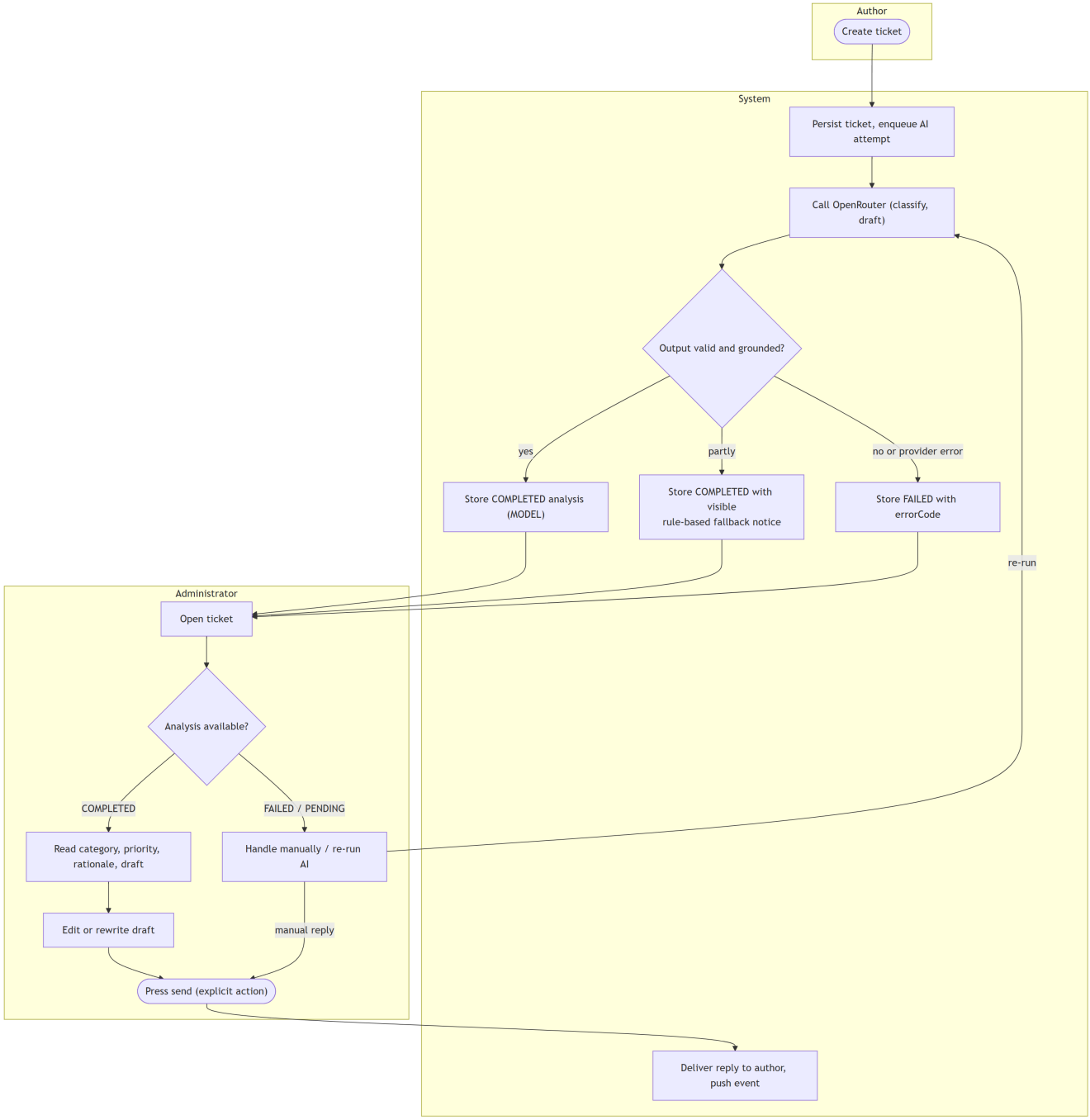
9. UML use cases



10. UML class diagram

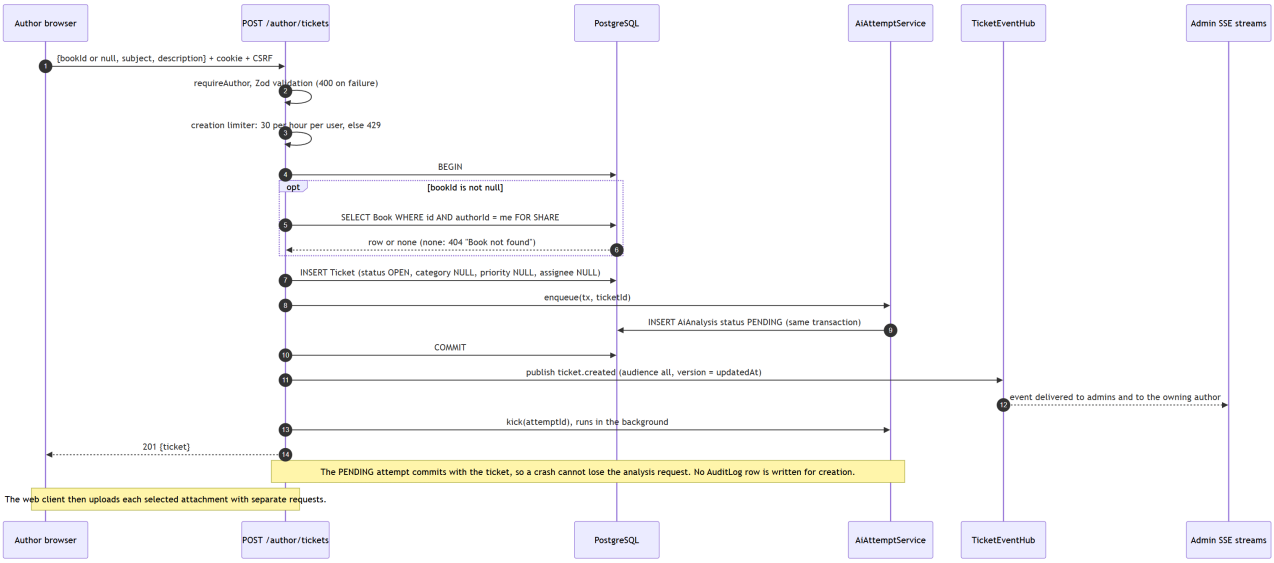
See §6.3. The domain classes mirror the Prisma models; there are no inheritance hierarchies in the implementation.

11. UML activity diagram

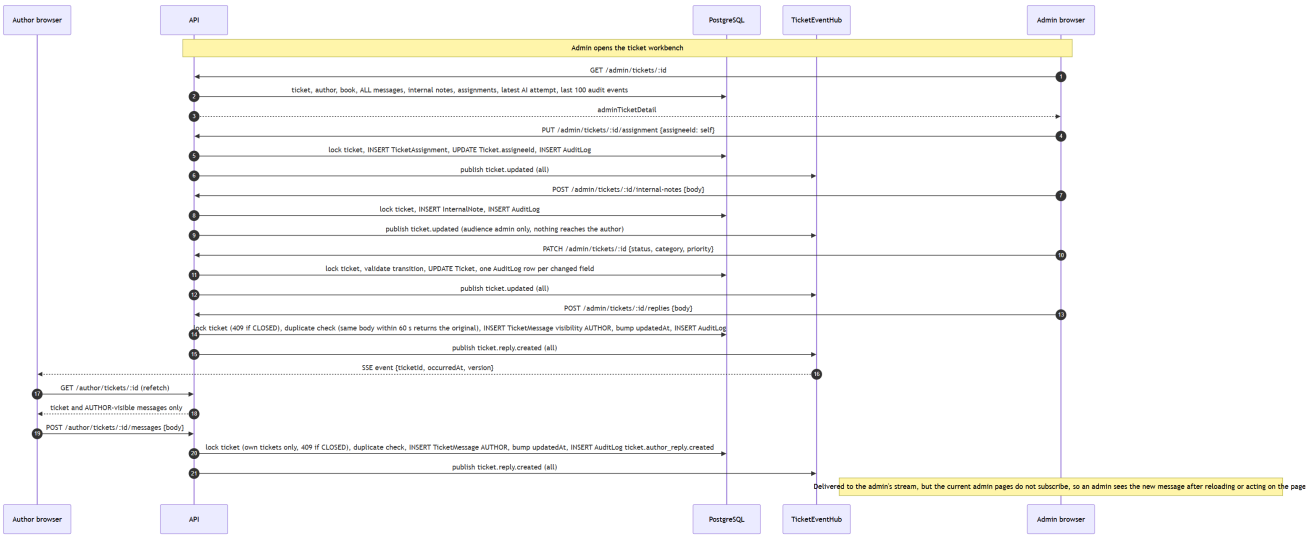


12. UML sequence diagrams

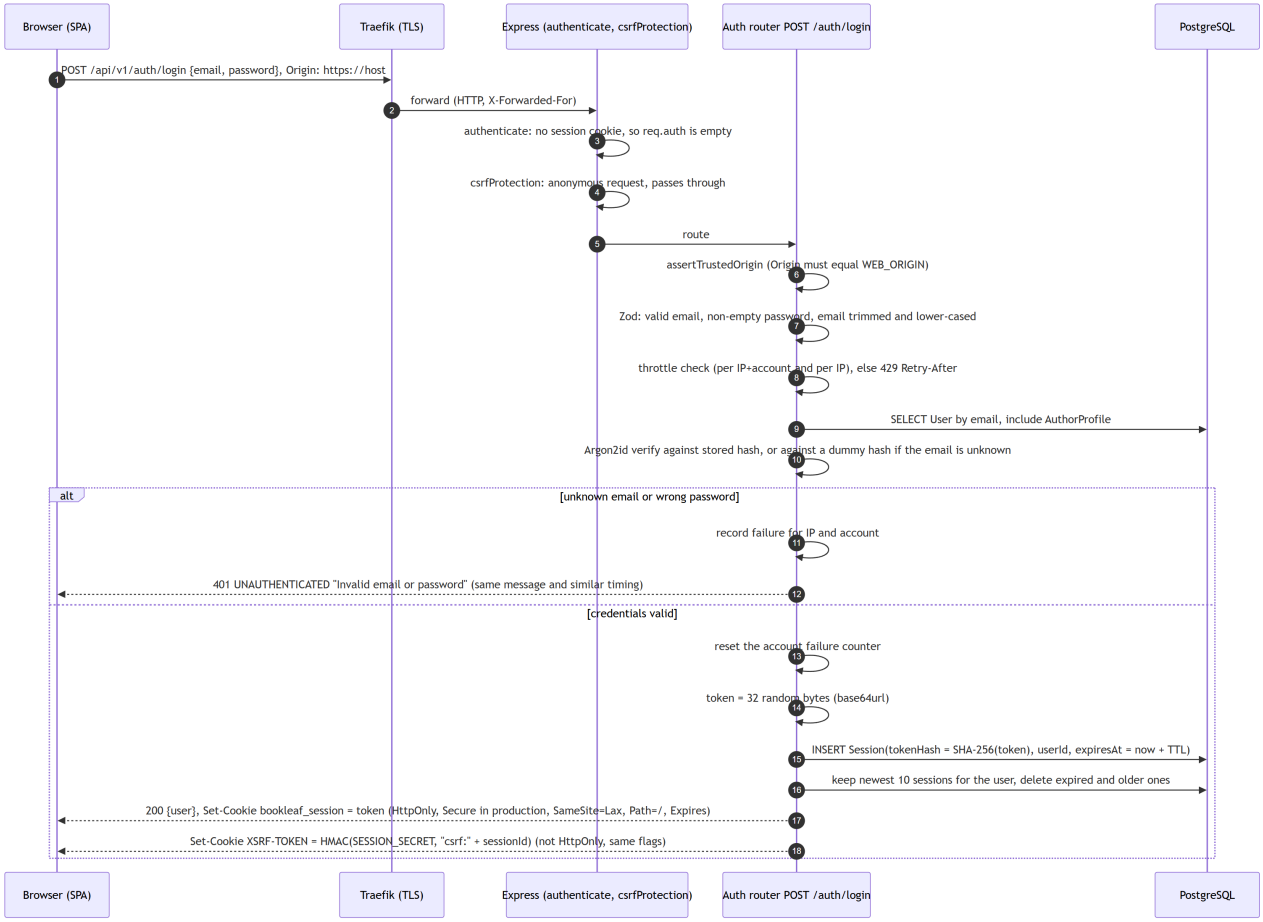
12.1 Ticket creation



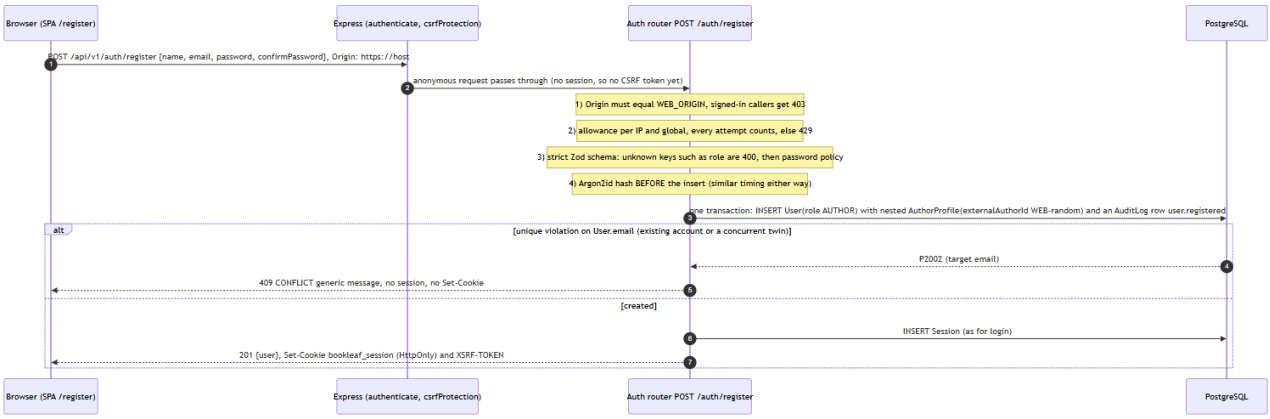
12.2 Conversation (reply, internal note, live event)



12.3 Login and session creation

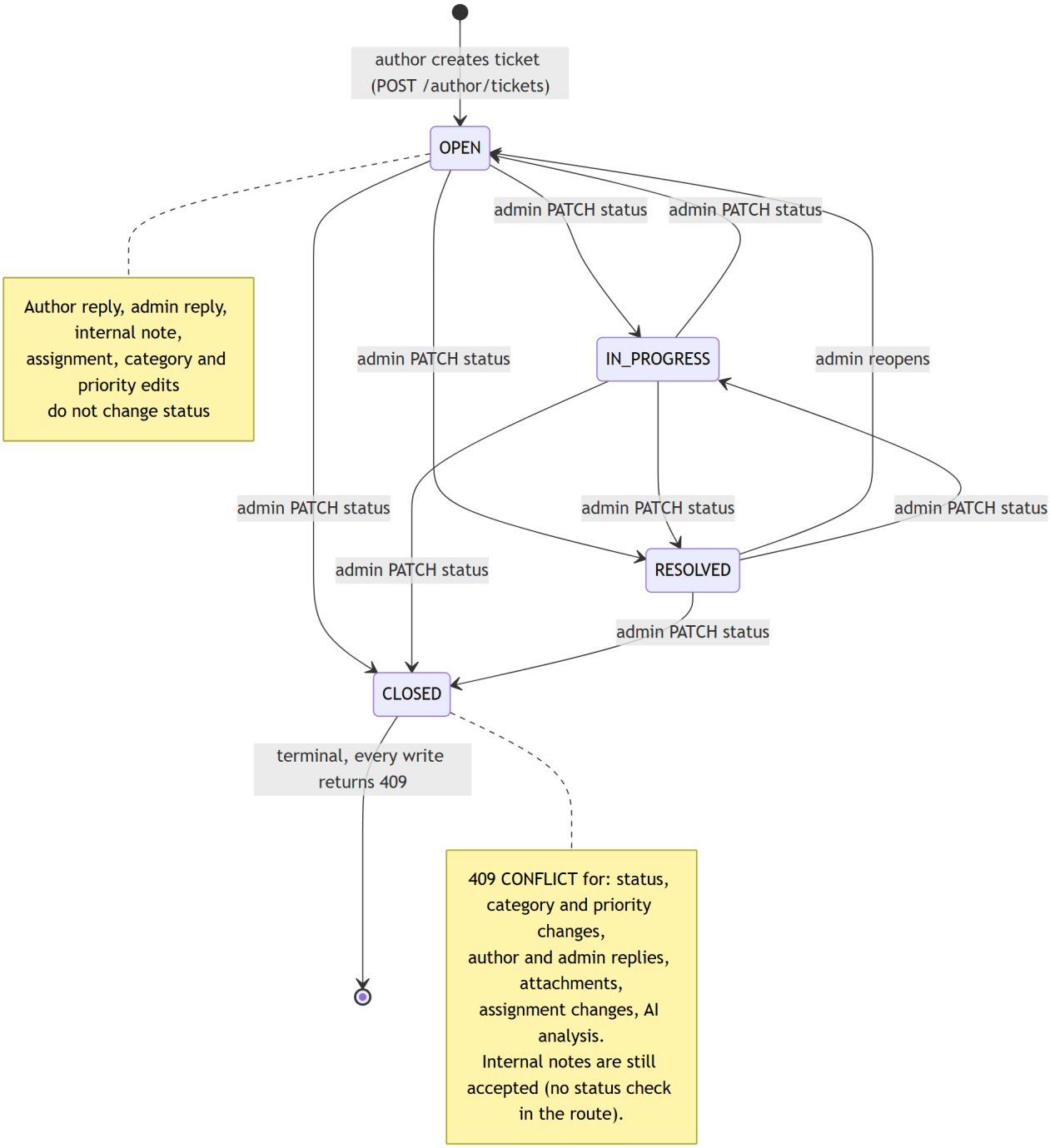


12.4 Author self-registration

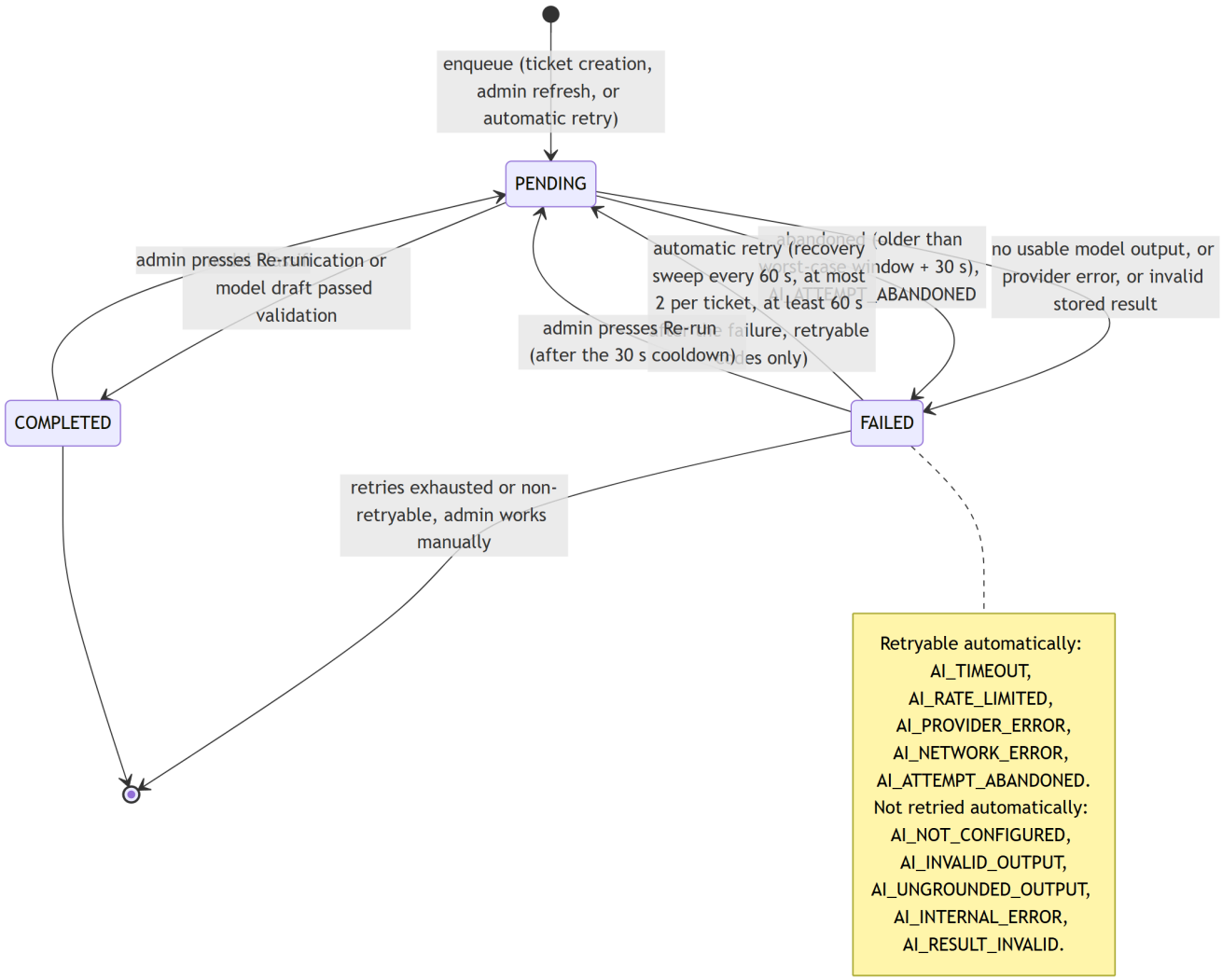


13. Ticket state transitions

Any non-CLOSED status can move to any other status through the admin `PATCH`. `CLOSED` is terminal: status, category, priority, replies, attachments, assignment and AI analysis answer `409`. Internal notes are still accepted on a closed ticket. Replies and notes never change status.



AI attempt states: PENDING, then exactly one of COMPLETED or FAILED. An attempt left PENDING beyond the worst-case window is closed as FAILED `AI_ATTEMPT_ABANDONED`.

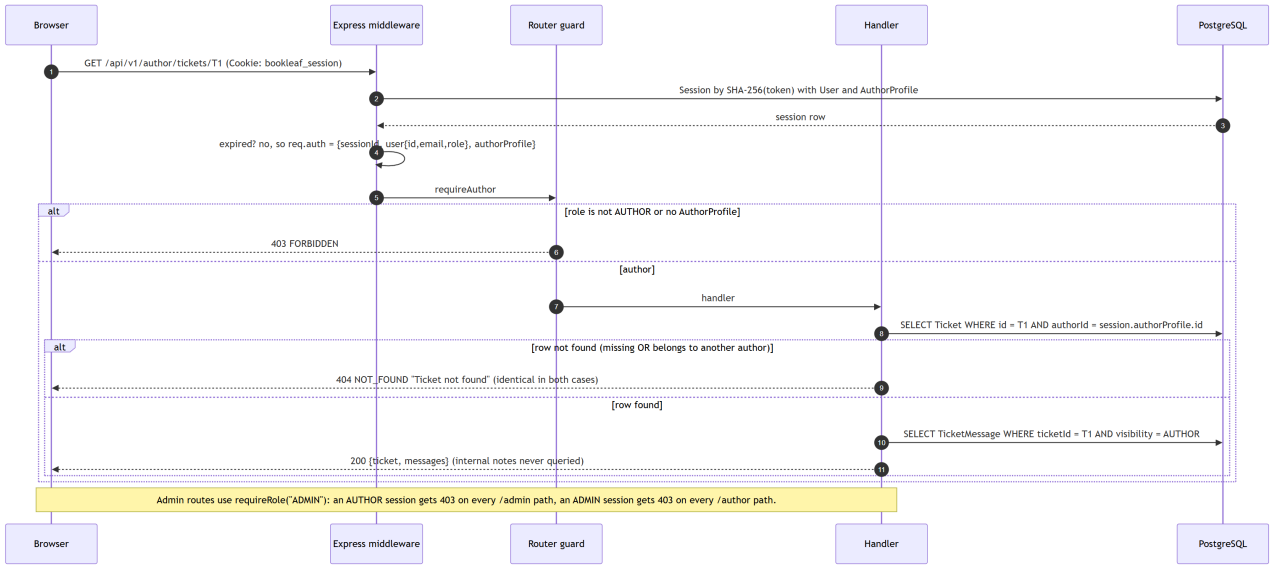


14. Authentication and authorisation

Model: server-side sessions with an opaque random cookie (`bookleaf_session`, HttpOnly, SameSite=Lax, Secure in production); only the SHA-256 of the token is stored. Passwords use Argon2id; unknown e-mails are verified against a dummy hash so timing and messages do not reveal accounts. A second cookie carries a session-bound CSRF token (an HMAC of the session id with `SESSION_SECRET`) that every state-changing request must echo in `X-CSRF-Token`, together with a trusted `Origin`. Login is throttled per IP and account. There is no JWT, OAuth or API-key authentication.

Registration and the fixed admin role. Authors register themselves at `/register` (`POST /auth/register`, §12.4). The endpoint accepts exactly four fields (name, e-mail, password, confirmation); any other field, including `role`, is rejected, and the server always stores `AUTHOR`. The password policy (12 to 128 characters with lowercase, uppercase and a digit, no very common passwords, no e-mail name) is shared by the form and the server, and passwords are hashed with the same Argon2id settings as login. Duplicates are decided only by the unique index on `User.email`, so concurrent registrations cannot create two accounts; the answer for an existing address is one generic `409`. Attempts are limited per IP (5 per hour) and globally (200 per hour). There is exactly one administrator, created and rotated only through the server-side tool `npm run admin:ensure` (credentials from the environment, never from a request, no default password, never promoting an author); no public API creates or modifies an admin.

Authorisation is role-based (`requireAuthor` , `requireRole(ADMIN)`) plus ownership scoping: an author's id comes only from the session and is part of every author query; a foreign resource returns the same `404` as a missing one. Admin-only data (internal notes, AI analysis) exists only on `/admin` routes.

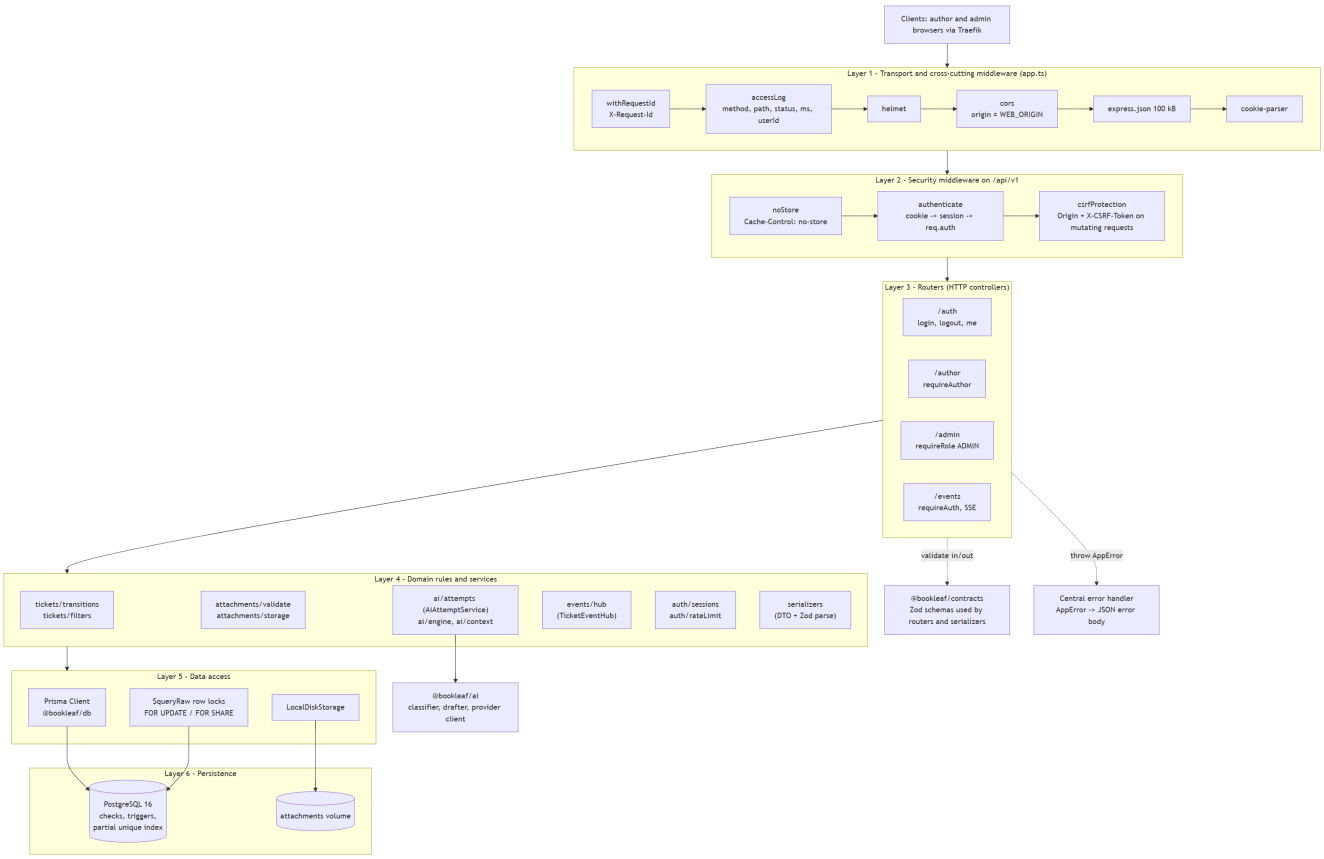


15. API architecture

Express application with a fixed middleware order (request id, security headers, JSON body limit 100 kB, authentication, CSRF, rate limits, validation, role guards, handlers) and Zod contracts shared with the frontend. Responses are JSON with `Cache-Control: no-store` ; errors use one envelope with stable codes. There are **25 operations**: 24 under `/api/v1` and `GET /healthz` .

Domain	Operations
Health	`GET /healthz` (liveness only)
Authentication	`POST /auth/login` , `POST /auth/register` , `POST /auth/logout` , `GET /auth/me`
Author books	`GET /author/books` , `GET /author/books/:bookId`
Author tickets	`POST /author/tickets` , `GET /author/tickets` , `GET /author/tickets/:ticketId` , `POST …/messages`
Author attachments	`POST` , `GET` list, `GET` one on `/author/tickets/:ticketId/attachments`
Admin tickets	`GET /admin/tickets` , `GET /admin/tickets/:ticketId` , `PATCH /admin/tickets/:ticketId`
Admin actions	`POST …/replies` , `POST …/internal-notes` , `PUT …/assignment`
AI	`GET /admin/tickets/:ticketId/ai-analysis` , `POST …/ai-analysis` (run or re-run)
Admin attachments	`GET` list, `GET` one
Events	`GET /events/tickets` (SSE)

Full request/response detail: `docs/architecture/10-API-REFERENCE.md` and `docs/architecture/openapi.yaml` .



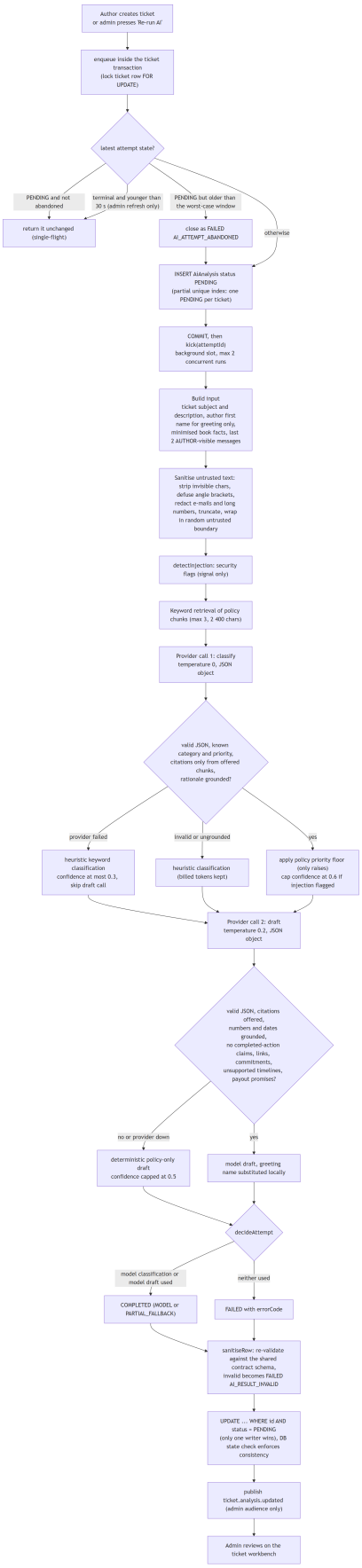
16. AI workflow (OpenRouter)

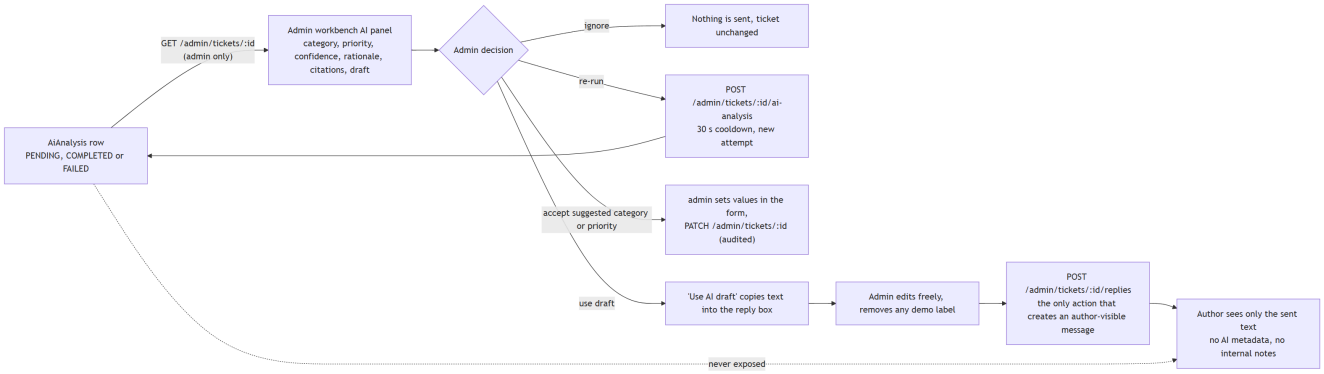
For each new ticket (and on admin request) the API creates a PENDING attempt in the ticket's transaction and runs it in the background (at most 2 concurrent). The `@bookleaf/ai` package then:

1. builds minimised input (subject, description, last two author-visible messages, minimised book facts; **no** e-mail, phone, name or financial figures; the author's first name is substituted locally after generation);
2. sanitises untrusted text (invisible characters, angle brackets, e-mail and long numbers redacted, random per-call boundary) and flags prompt-injection signals;
3. retrieves up to three policy chunks from a static knowledge base (keyword retrieval, no embeddings);
4. calls the model to **classify** (JSON), then to **draft** (JSON);
5. validates both outputs against schemas, cited policy ids and a grounding check (every number/date must come from the sources; no completed-action claims, links, refund promises, unsupported timeframes, payout promises, invented quoted menu/section names, or telling the author to reply to an e-mail);
6. stores the result: COMPLETED (model), COMPLETED with a visible *rule-based fallback* notice in the rationale when part of the model output was unusable, or FAILED with a stable error code (`AI_RATE_LIMITED`, `AI_TIMEOUT`, `AI_PROVIDER_ERROR`, `AI_NETWORK_ERROR`, `AI_INVALID_OUTPUT`, `AI_UNGROUNDED_OUTPUT`, `AI_NOT_CONFIGURED`, …).

Model output is treated as untrusted data. The package has no send capability; `requiresHumanReview` is the literal `true`.

Zero-spend controls: the default model is `nvidia/nemotron-3-super-120b-a12b:free` (`google/gemma-4-31b-it:free` is a supported alternative); against a real endpoint, configuration is rejected unless the primary and any fallback model ids end in `:free`; no paid fallback exists; retries are bounded (default 2) with jittered backoff and `Retry-After` respected; timeout 15 s; `max_tokens` capped (default 400); reasoning is disabled in requests so hidden reasoning tokens cannot exhaust the cap.


Author creates ticket
or admin presses 'Re-run AI'
enqueue inside the ticket transaction
(lock ticket row FOR UPDATE)
latest attempt state?
PENDING and not abandoned
terminal and younger than 30 s (admin refresh only)
PENDING but older than the worst-case window
otherwise
return it unchanged (single-flight)
close as FAILED AI_ATTEMPT_ABANDONED
INSERT AIAnalysis status PENDING (partial unique index: one PENDING per ticket)
COMMIT, then kick(attemptId) background slot, max 2 concurrent runs
Build input ticket subject and description, author first name for greeting only, minimised book facts, last 2 AUTHOR-visible messages
Sanitise untrusted text: strip invisible chars, defuse angle brackets, redact e-mails and long numbers, truncate, wrap in random untrusted boundary
detectInjection: security flags (signal only)
Keyword retrieval of policy chunks (max 3, 2 400 chars)
Provider call 1: classify temperature 0, JSON object
valid JSON, known category and priority, citations only from offered chunks, rationale grounded?
provider failed
invalid or ungrounded
yes
heuristic keyword classification confidence at most 0.3, skip draft call
heuristic classification (billed tokens kept)
apply policy priority floor (only raises) cap confidence at 0.6 if injection flagged
Provider call 2: draft temperature 0.2, JSON object
valid JSON, citations offered, numbers and dates grounded, no completed-action claims, links, commitments, unsupported timelines, payout promises?
no or provider down
yes
deterministic policy-only draft confidence capped at 0.5
model draft, greeting name substituted locally
decideAttempt
model classification or model draft used
neither used
COMPLETED (MODEL or PARTIAL_FALLBACK)
FAILED with errorCode
sanitiseRow: re-validate against the shared contract schema, invalid becomes FAILED AI_RESULT_INVALID
UPDATE ... WHERE id AND status = PENDING (only one writer wins), DB state check enforces consistency
publish ticket.analysis.updated (admin audience only)
Admin reviews on the ticket workbench



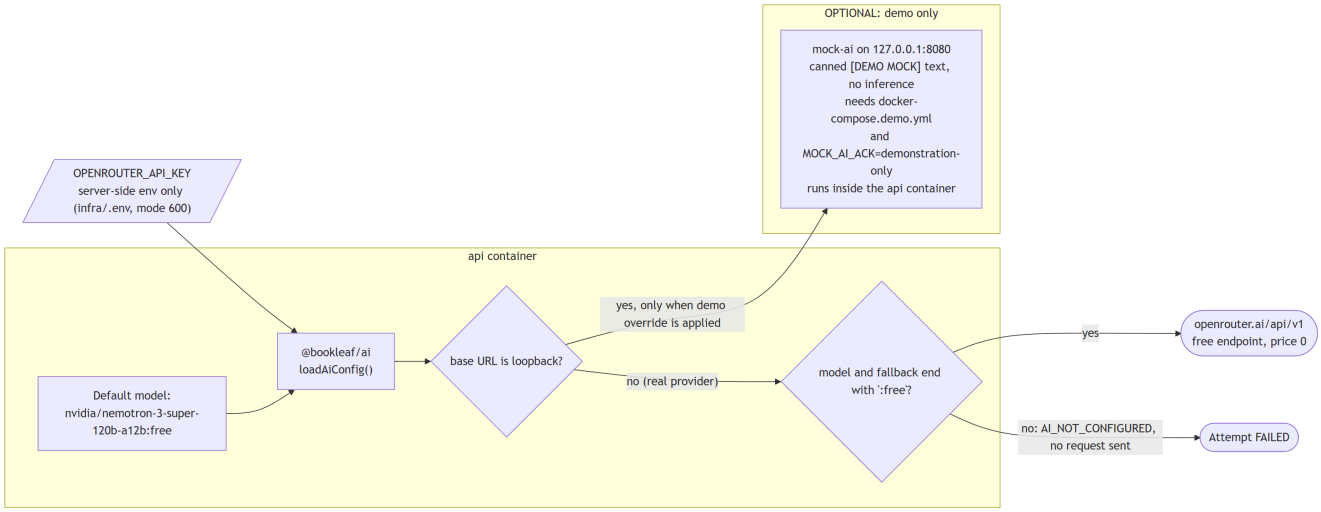
Observed with a real free model (2026-10-16, synthetic tickets, account spend 0). On the default `nvidia/nemotron-3-super-120b-a12b:free` with the final prompt: classification was correct for all six categories in every run, and the draft passed the strict validator in 12 of 13 analyses (it was 3 of 6 before prompt hardening), at 4-14 s per analysis and reported cost 0. The failures that remained were a model promising an unverifiable time frame or inventing a product feature, which the validator rejects and replaces with a labelled rule-based draft. The sample is small and non-deterministic. The Gemma alternative returned HTTP 429 from its upstream provider throughout and is unverified. Real failure handling was exercised against the live endpoint: rejected key (1 request, no retry), nonexistent model, paid-model configuration (0 requests) and a rate limit; none produced mock or paid output. Details, history and caveats: `docs/release/AI_REAL_PROVIDER.md`.

Draft-quality hardening. The knowledge base's tone guideline and royalty style example contain illustrative promises such as "within 48 hours", which real models copied and the grounding check rejected. The knowledge-base text was **not** rewritten, because it could not be verified as an operational commitment; the prompt now labels those blocks as style-only illustrations, masks durations inside them (policy excerpts are never masked), and tells the model what to do when facts are missing (acknowledge, say what the team will verify, commit to no dates, amounts, response times or distribution).

17. Mock versus real provider modes

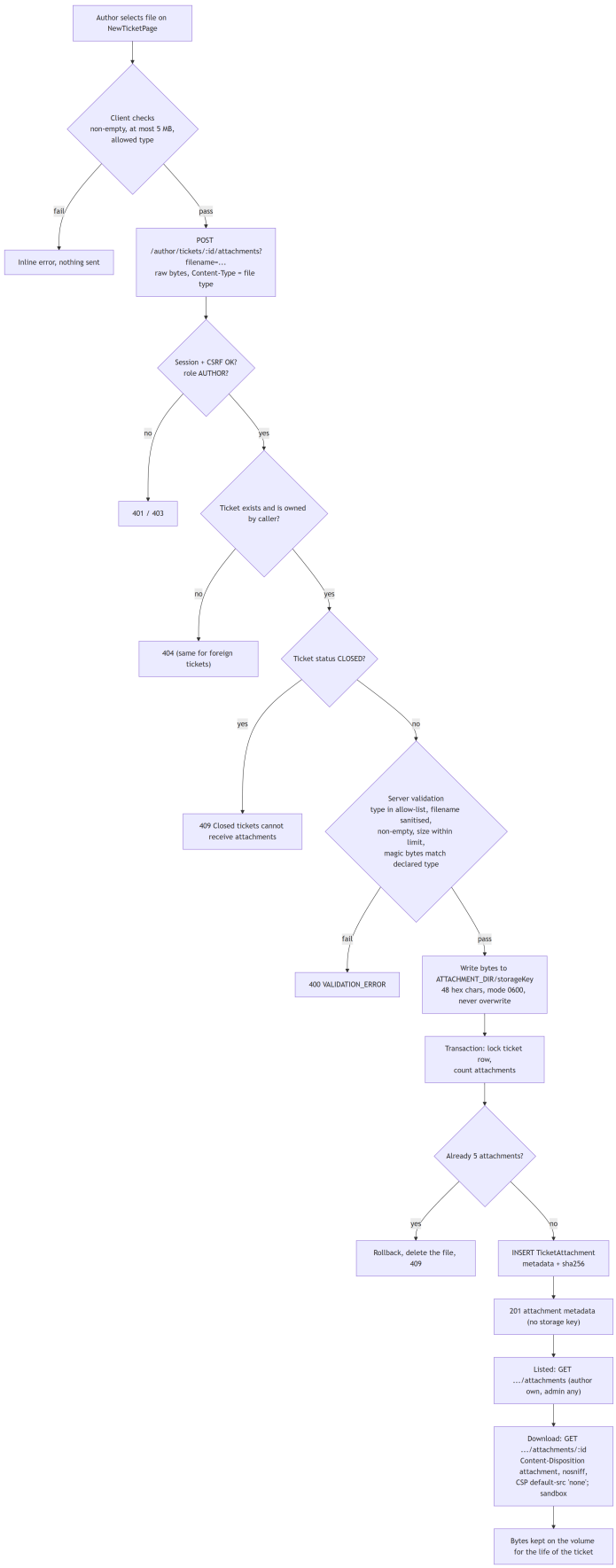
	Real provider (default)	**Demonstration mock (opt-in)**
Enabled by	`OPENROUTER_API_KEY` in `infra/.env`; base URL left at the default	`infra/docker-compose.demo.yml` **and** `MOCK_AI_ACK=demonstration-only`
Endpoint	`https://openrouter.ai/api/v1`	`http://127.0.0.1:8080` inside the api container
Output	Language-model text	Canned, keyword-matched text labelled `[DEMO MOCK]` / `[DEMO DRAFT]`; no inference
Model guard	`:free` ids only	exempt (no billable calls)
On failure	Attempt FAILED with an error code	n/a

The system never substitutes mock output for a failed real call. Since 2026-10-16 the staging site runs the **real provider** (mock disabled); the demonstration mock remains an opt-in override only.



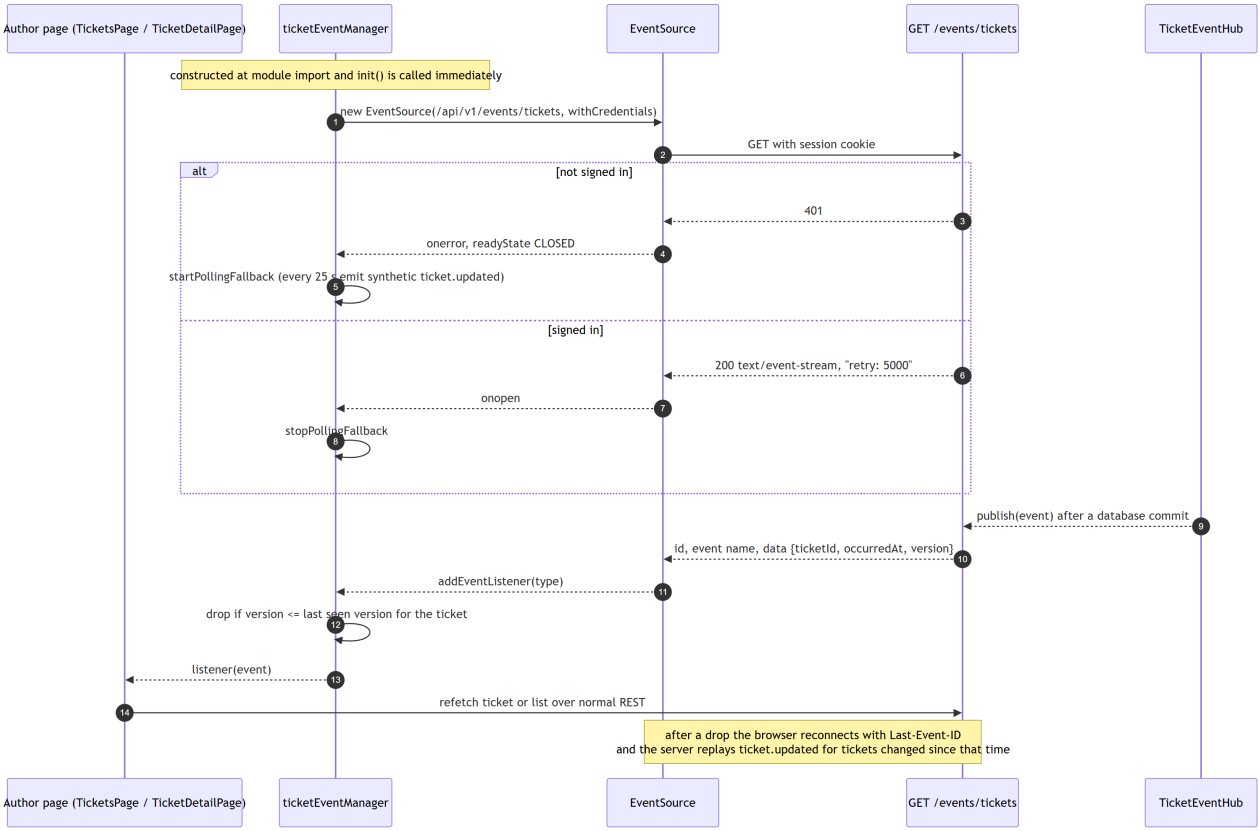
18. Attachment handling

Authors upload files to their own tickets (client and server check: non-empty, at most 5 MB, type in an allow-list, magic bytes matching the declared type, sanitised filename). Files are stored on a Docker volume under random keys with a SHA-256; metadata is in `TicketAttachment`. Downloads are forced (`Content-Disposition: attachment`, `nosniff`, a restrictive CSP) and authorised by ownership (author) or role (admin). Deletion is not implemented.


Author selects file on NewTicketPage
Client checks non-empty, at most 5 MB, allowed type
fail
pass
Inline error, nothing sent
POST /author/tickets/:id/attachments?filename=... raw bytes, Content-Type = file type
Session + CSRF OK? role AUTHOR?
no
yes
401 / 403
TIcket exists and is owned by caller?
no
yes
404 (same for foreign tickets)
Ticket status CLOSED?
yes
no
409 Closed tickets cannot receive attachments
Server validation type in allow-list, filename sanitised, non-empty, size within limit, magic bytes match declared type
fail
pass
400 VALIDATION_ERROR
Write bytes to ATTACHMENT_DIR/storageKey 48 hex chars, mode 0600, never overwrite
Transaction: lock ticket row, count attachments
Already 5 attachments?
yes
no
Rollback, delete the file, 409
INSERT TicketAttachment metadata + sha256
201 attachment metadata (no storage key)
Listed: GET .../attachments (author own, admin any)
Download: GET .../attachments/:id Content-Disposition attachment, nosniff, CSP default-src 'none'; sandbox
Bytes kept on the volume for the life of the ticket

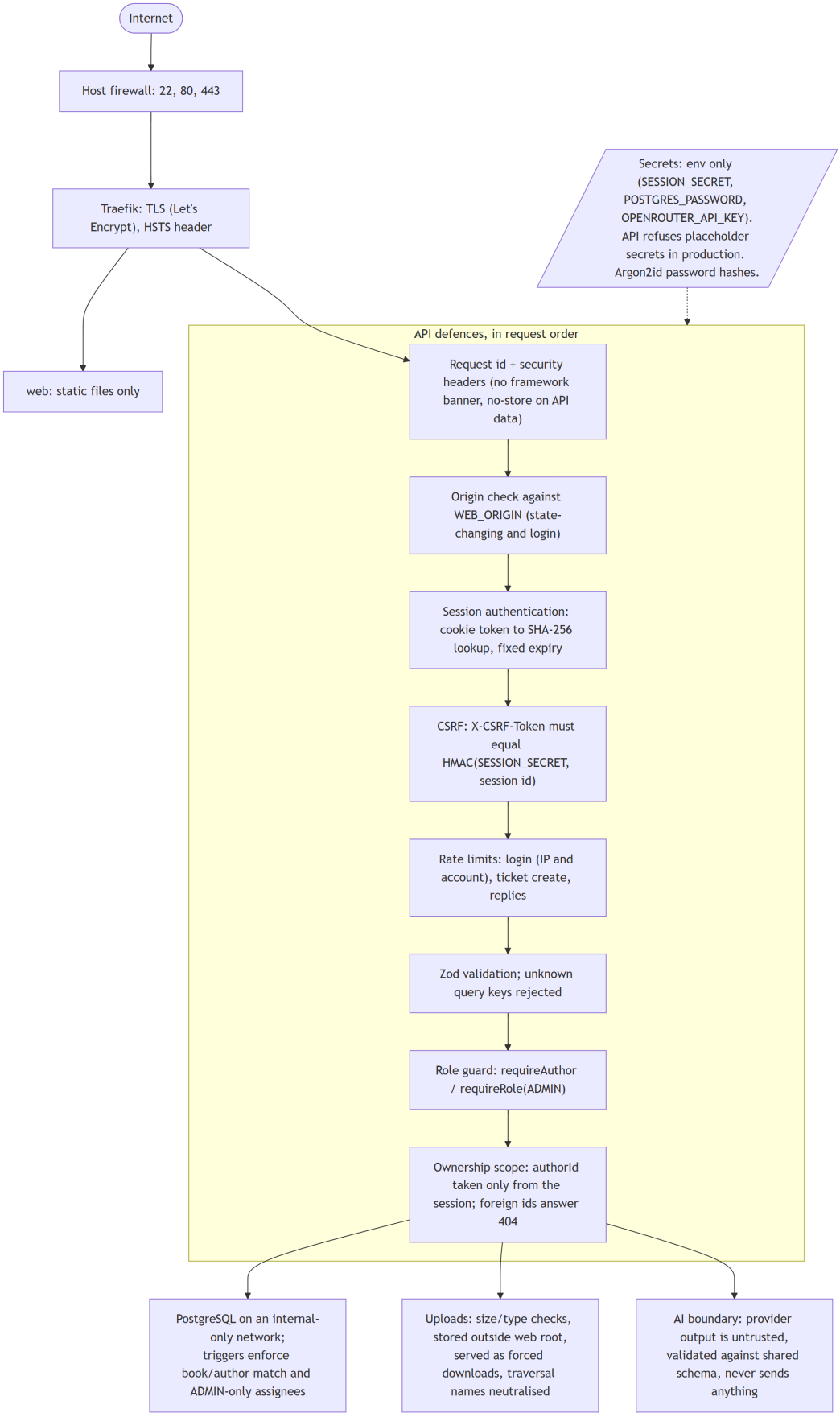
19. Real-time updates: SSE and polling

`GET /events/tickets` is a Server-Sent Events stream. An in-process hub fans out events after each change; admins receive everything, authors only events for their own tickets, and AI and internal-note events are admin-only. The browser reconnects with `Last-Event-ID` and the server replays tickets changed since then; the stream asks for a 5 s retry. If the stream is unavailable the client falls back to a 25 s poll and stops polling when SSE recovers. The hub is per process (single API instance).



20. Security design

Layers: host firewall (22, 80, 443) → Traefik TLS and HSTS → API origin check → session authentication → CSRF → rate limits (login; registration; ticket creation 30 per hour per user; replies) → Zod validation (unknown query keys rejected) → role guard → ownership scoping → database constraints and triggers. Secrets exist only in server environment files; the API refuses placeholder secrets in production; the OpenRouter key never reaches the browser, logs, database rows or API responses (verified by tests with a fake key).



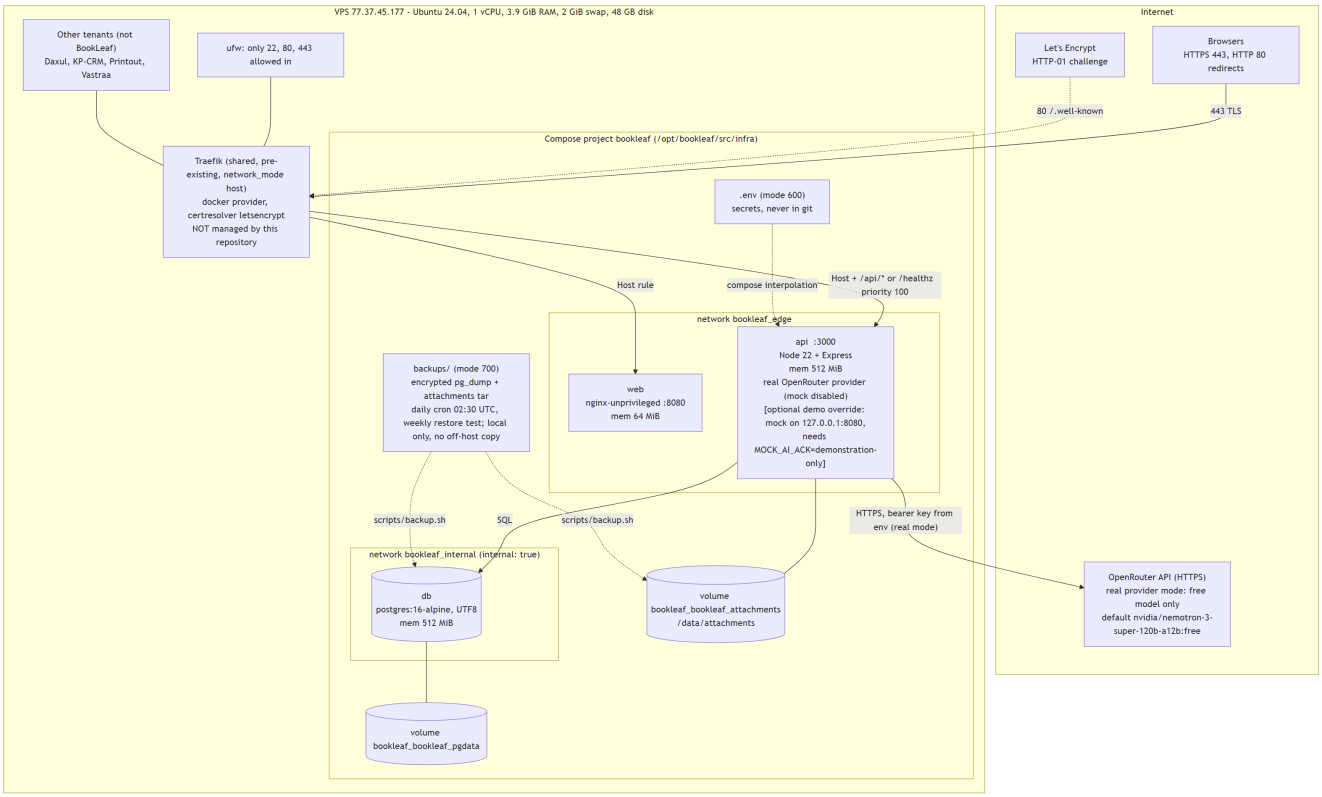
Residual risks: free AI endpoints may retain prompts (only synthetic data was used in testing); login and registration throttling is in memory; one API instance; registration reveals whether an address is already registered (no e-mail verification); self-registered authors are not verified people, so book linkage must stay a manual administrative decision.

Data hygiene controls: the production seed refuses synthetic accounts (`NODE_ENV=production` without an explicit acknowledgement), and a dry-run-by-default cleanup tool removes only provably synthetic records, never the admin and never audit history (`docs/release/FINAL_CLEANUP_INVENTORY.md`).

21. Docker and VPS deployment

Compose project `bookleaf` with services `db` (postgres:16-alpine, internal network only), `api` and `web`, plus one-shot `migrate` and `seed` jobs. Traefik (shared, outside this repository) terminates TLS and routes the host and `/api/` and `/healthz`. Secrets come from `infra/.env` (mode 600, git-ignored). Read-only inspection on 2026-10-16 found all BookLeaf containers healthy, TLS valid until 2027-01-13, disk 30% used, and only ports 22/80/443 exposed.

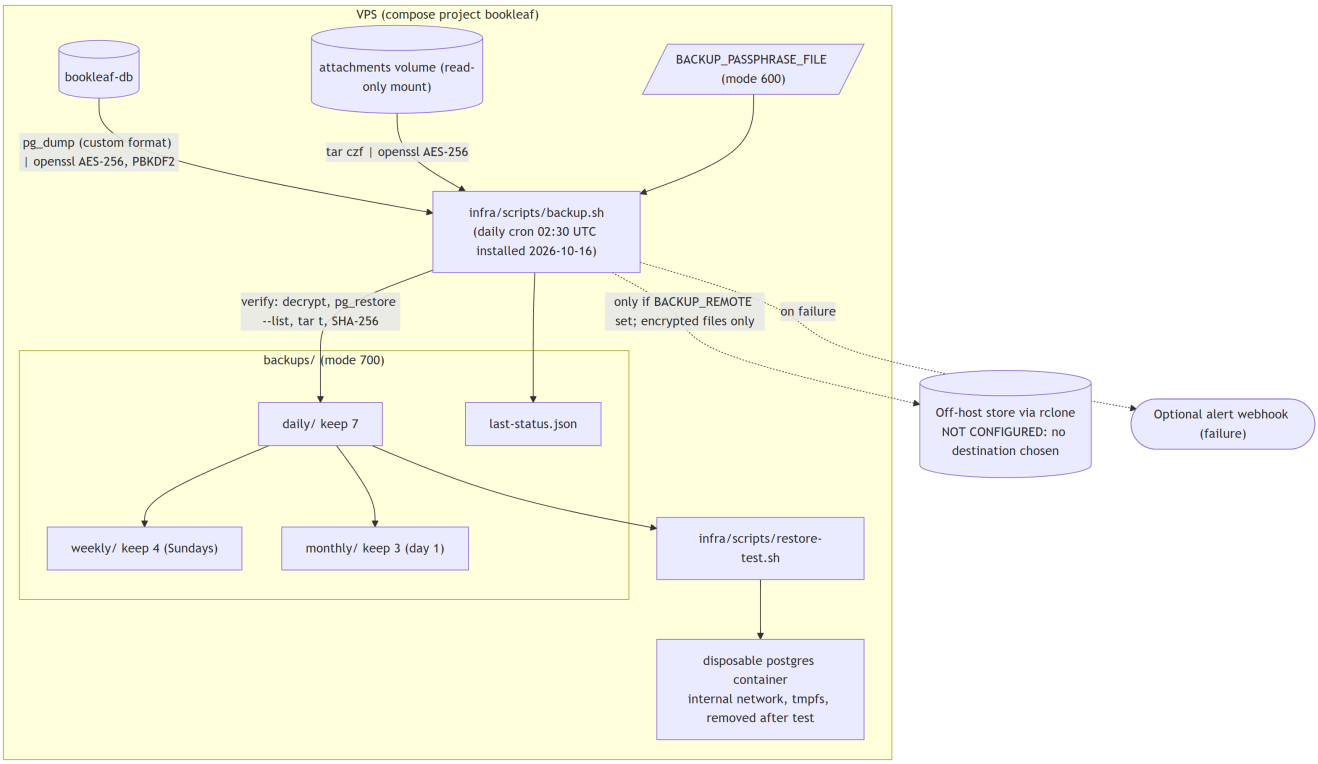
Deployment findings to act on: a host cron prunes unused Docker images older than 24 h (can delete rollback images); the deployed source tree is not a git checkout; a reboot is pending. Deployment procedures and the prepared (not executed) release plan are in `infra/DEPLOY.md` and `docs/release/`.



22. Backup and recovery

`infra/scripts/backup.sh` writes an encrypted (AES-256, PBKDF2) custom-format database dump and an attachments archive, verifies them (decrypt, `pg_restore --list`, `tar t`, SHA-256 manifest), rotates 7 daily / 4 weekly / 3 monthly sets, records `last-status.json`, can alert a webhook and can upload to an rclone destination. `restore-test.sh` restores into a disposable container. **Status:** tested locally with disposable containers only (550-row restore, wrong passphrase and tampered archive rejected, retention). Installed on the VPS (daily 02:30 UTC cron, weekly restore test) on 2026-10-16; a restore of a backup of the live database was proven in a disposable container (119 tickets, 25 attachments, 122 AI attempts matched). No

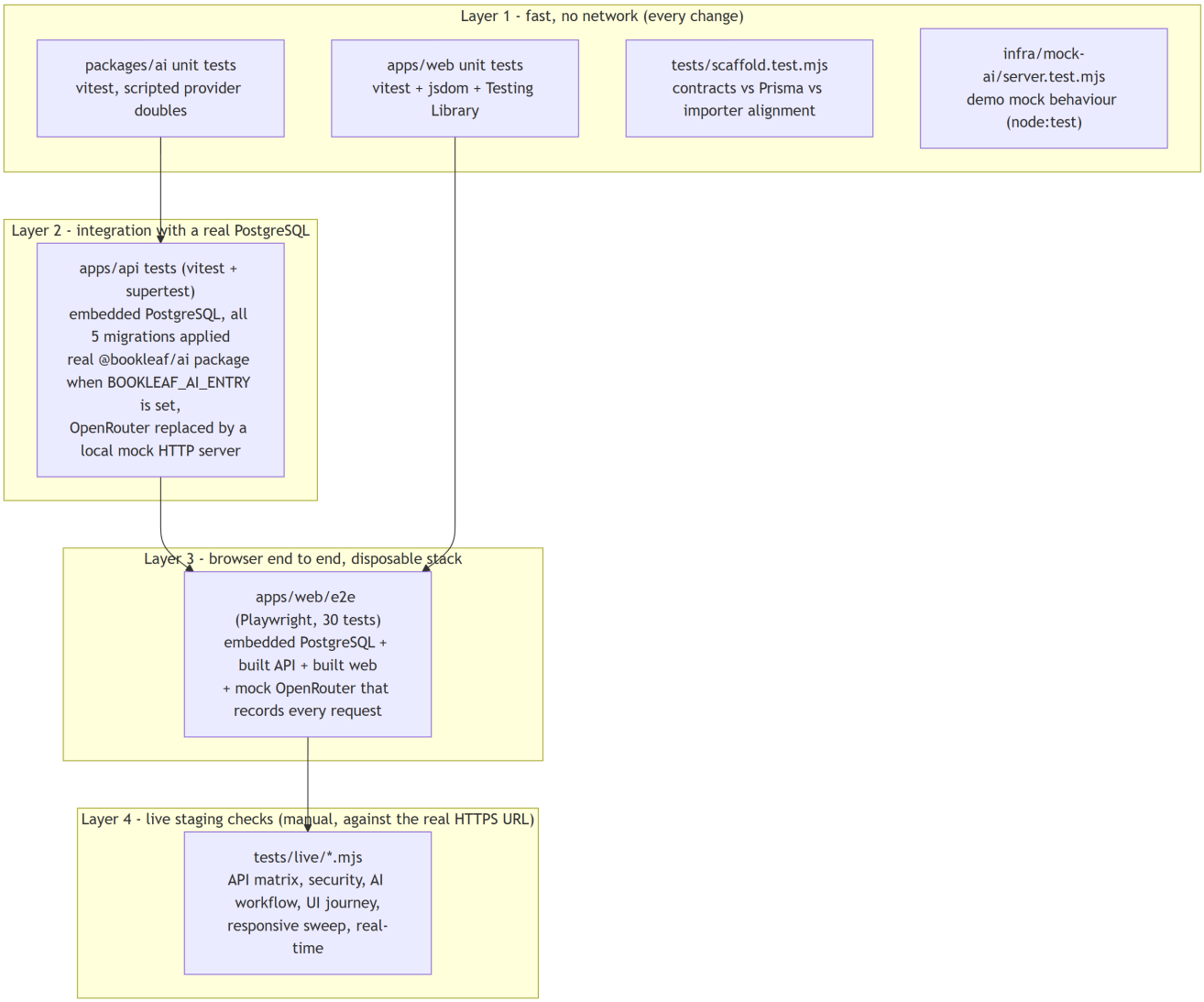
off-host destination exists, so backups are local-only; the first scheduled run is not yet observed. Recovery objectives are targets, not guarantees. See `docs/operations/BACKUP_AND_RECOVERY.md` and `DISASTER_RECOVERY.md`.



23. Testing strategy and results

Layer	Tool	Result (this branch)
AI package	Vitest	485 passed
API with real PostgreSQL	Vitest + Supertest	292 passed, 0 skipped (42 registration, 20 cleanup/seed-guard/admin-tool)
Frontend	Vitest + Testing Library	68 passed (15 registration form)
Browser E2E	Playwright (Chrome) on a disposable stack with a mock provider	41 passed (11 registration: desktop, 375 px mobile, keyboard)
Scaffold / infra	`node --test`	9 / 12 passed
Backup scripts	manual, disposable containers	passed (see §22)
Real OpenRouter	`tests/live/ai-real.mjs`, `ai-real-failures.mjs`	classification 6/6; drafts accepted 12/13 on the final prompt; failure scenarios passed (see §16)
Live production checks	`tests/live/*` and ad-hoc scripts, 2026-10-16	10/10 original-author logins with exact book ids, cross-author read 404, author blocked from admin API, self-registration, ticket, admin note, override and reply, author receipt, internal note hidden (13/13); no AI calls (quota exhausted)

Regression tests were added for each fix made in this release (whitespace login e-mail, signed-in redirect from `/login`, fallback labelling, model-greeting handling, zero-spend guard). All automated AI tests use mock or scripted providers.



24. Known limitations

- Real AI is verified with limits: small, non-deterministic sample (12 of 13 drafts accepted); the Gemma alternative was rate limited upstream; free models rate-limit (20/min, 50/day documented without credits) and may retain prompts; the validator cannot judge semantic quality, so human review is essential.
- The knowledge base's tone guideline and royalty example suggest quoting "within 48 hours"; this is unverified as a real commitment, the AI is told not to repeat it, and the policy text was not changed. If it is a real commitment it should be added to a policy chunk.
- Real AI on the live site is unverified because the free daily quota (50 requests) was exhausted during acceptance; a key without purchased credits supports about 25 analyses a day.
- Backups are installed and scheduled, with a restore proven in isolation, but there is no off-host copy, so disaster recovery is incomplete; the first scheduled run is not yet observed.
- Host risks: image-prune cron, pending reboot and updates, deployed tree not a git checkout.

- Deferred defects: duplicate tickets on concurrent identical submissions; admin search covers the current page only; health endpoint is liveness-only; internal-note/assignment deletion not trigger-protected (needs a migration).
- Not implemented: password reset, e-mail notifications, e-mail verification of new accounts, attachment deletion, server-side search.
- Self-registered authors start with no books: there is no admin screen or API to link books to an account yet (done by a database operation), and registration reveals whether an address is already registered.
- Registration is live and was exercised on the production site; the production database still holds older synthetic test records (inventory: `docs/release/CLEANUP_INVENTORY_2026-10-16.md`) until a separately approved cleanup is run.
- Single API instance: login throttling, event hub and AI single-flight are per process.

25. Conclusion

The portal implements the full author-to-admin support workflow with strong tenant isolation, a defence-in-depth security model, and an AI assistant that is advisory only, grounded in policy and fail-closed. Local automated verification is extensive and green. Remaining work before calling it production-ready: repeat real-provider verification at larger scale and on the deployed site, install and prove the backup chain with an off-host copy, secure the rollback images, and run the planned deployment.